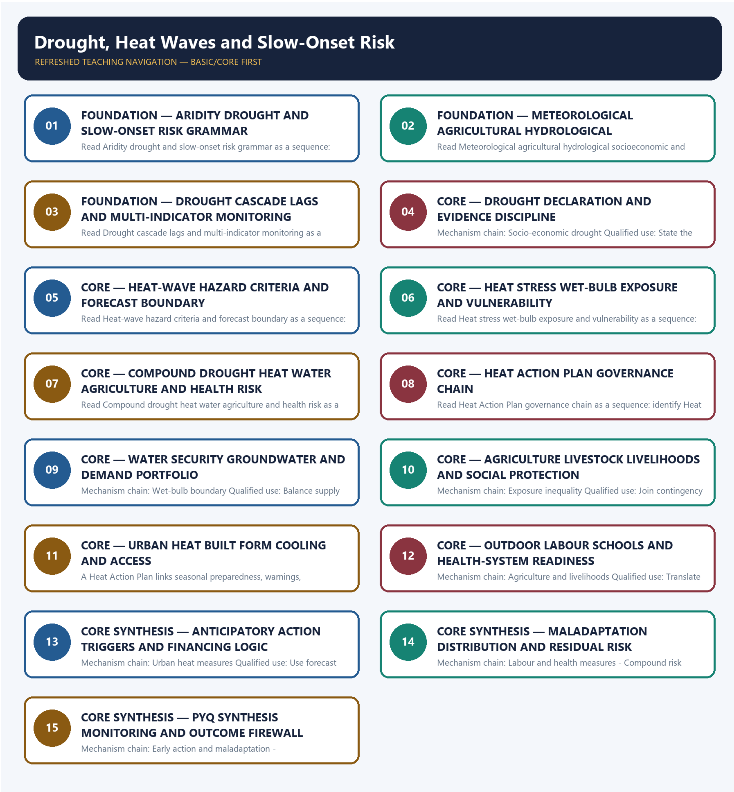
SOLVED PRACTICE WORKBOOK

Drought, Heat Waves and Slow-Onset Risk - Solved Practice Workbook

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing



Original deterministic study visual; labels are explanatory, not textual quotations.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

CONTENTS / WORKBOOK INDEX

Major learning parts and meaningful subtopics are listed in document order. Page numbers are generated from the final PDF layout; PDF bookmarks mirror the same hierarchy.

Drought, Heat Waves and Slow-Onset Risk - Solved Practice Workbook

BASIC MCQS / REMEDIATION

Q1. Which statement correctly identifies Aridity versus drought?

A. Aridity is a long-term climatic condition of low moisture availability, whereas drought is a temporary or episodic deficit relative to the expected local water regime. B. Meteorological drought concerns prolonged inadequate or poorly distributed precipitation relative to the relevant climatology; it is a hazard indicator, not a complete account of impacts. C. Agricultural drought arises when soil moisture and crop-water availability become inadequate for crops, so rainfall information must be read with sowing, vegetation and soil-moisture evidence. D. Hydrological drought concerns below-normal availability in rivers, reservoirs, lakes or aquifers and may lag behind the rainfall deficit that initiated it.

Answer: A. Explanation: Aridity is a long-term climatic condition of low moisture availability, whereas drought is a temporary or episodic deficit relative to the expected local water regime. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q2. Which option preserves the risk or institutional boundary of Aridity versus drought?

A. Socio-economic drought occurs when water shortage disrupts the supply of goods, services, livelihoods or essential uses; it expresses the interaction of physical scarcity with demand and unequal access. B. Aridity is a long-term climatic condition of low moisture availability, whereas drought is a temporary or episodic deficit relative to the expected local water regime. C. Hydrological drought concerns below-normal availability in rivers, reservoirs, lakes or aquifers and may lag behind the rainfall deficit that initiated it. D. Agricultural drought arises when soil moisture and crop-water availability become inadequate for crops, so rainfall information must be read with sowing, vegetation and soil-moisture evidence.

Answer: B. Explanation: Aridity is a long-term climatic condition of low moisture availability, whereas drought is a temporary or episodic deficit relative to the expected local water regime. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q3. Which statement uses Aridity versus drought without changing its hazard, mandate or status?

A. Socio-economic drought occurs when water shortage disrupts the supply of goods, services, livelihoods or essential uses; it expresses the interaction of physical scarcity with demand and unequal access. B. Ecological drought describes water shortage severe enough to impair ecosystem productivity or function; it must not be silently substituted for socio-economic drought. C. Aridity is a long-term climatic condition of low moisture availability, whereas drought is a temporary or episodic deficit relative to the expected local water regime. D. Hydrological drought concerns below-normal availability in rivers, reservoirs, lakes or aquifers and may lag behind the rainfall deficit that initiated it.

Answer: C. Explanation: Aridity is a long-term climatic condition of low moisture availability, whereas drought is a temporary or episodic deficit relative to the expected local water regime. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q4. Which option avoids the standard UPSC close-option trap about Aridity versus drought?

A. Drought can unfold from rainfall deficit through soil-moisture stress, storage decline, ecosystem damage, livelihood loss and health effects over different time scales, so one indicator cannot represent every stage. B. Ecological drought describes water shortage severe enough to impair ecosystem productivity or function; it must not be silently substituted for socio-economic drought. C. Socio-economic drought occurs when water shortage disrupts the supply of goods, services, livelihoods or essential uses; it expresses the interaction of physical scarcity with demand and unequal access. D. Aridity is a long-term climatic condition of low moisture availability, whereas drought is a temporary or episodic deficit relative to the expected local water regime.

Answer: D. Explanation: Aridity is a long-term climatic condition of low moisture availability, whereas drought is a temporary or episodic deficit relative to the expected local water regime. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q5. Which statement correctly identifies Meteorological drought?

A. Meteorological drought concerns prolonged inadequate or poorly distributed precipitation relative to the relevant climatology; it is a hazard indicator, not a complete account of impacts. B. Hydrological drought concerns below-normal availability in rivers, reservoirs, lakes or aquifers and may lag behind the rainfall deficit that initiated it. C. Socio-economic drought occurs when water shortage disrupts the supply of goods, services, livelihoods or essential uses; it expresses the interaction of physical scarcity with demand and unequal access. D. Agricultural drought arises when soil moisture and crop-water availability become inadequate for crops, so rainfall information must be read with sowing, vegetation and soil-moisture evidence.

Answer: A. Explanation: Meteorological drought concerns prolonged inadequate or poorly distributed precipitation relative to the relevant climatology; it is a hazard indicator, not a complete account of impacts. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q6. Which option preserves the risk or institutional boundary of Meteorological drought?

A. Ecological drought describes water shortage severe enough to impair ecosystem productivity or function; it must not be silently substituted for socio-economic drought. B. Meteorological drought concerns prolonged inadequate or poorly distributed precipitation relative to the relevant climatology; it is a hazard indicator, not a complete account of impacts. C. Hydrological drought concerns below-normal availability in rivers, reservoirs, lakes or aquifers and may lag behind the rainfall deficit that initiated it. D. Socio-economic drought occurs when water shortage disrupts the supply of goods, services, livelihoods or essential uses; it expresses the interaction of physical scarcity with demand and unequal access.

Answer: B. Explanation: Meteorological drought concerns prolonged inadequate or poorly distributed precipitation relative to the relevant climatology; it is a hazard indicator, not a complete account of impacts. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q7. Which statement uses Meteorological drought without changing its hazard, mandate or status?

A. Ecological drought describes water shortage severe enough to impair ecosystem productivity or function; it must not be silently substituted for socio-economic drought. B. Socio-economic drought occurs when water shortage disrupts the supply of goods, services, livelihoods or essential uses; it expresses the interaction of physical scarcity with demand and unequal access. C. Meteorological drought concerns prolonged inadequate or poorly distributed precipitation relative to the relevant climatology; it is a hazard indicator, not a complete account of impacts. D. Drought can unfold from rainfall deficit through soil-moisture stress, storage decline, ecosystem damage, livelihood loss and health effects over different time scales, so one indicator cannot represent every stage.

Answer: C. Explanation: Meteorological drought concerns prolonged inadequate or poorly distributed precipitation relative to the relevant climatology; it is a hazard indicator, not a complete account of impacts. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage,

evidence rung or implementation status.

Q8. Which option avoids the standard UPSC close-option trap about Meteorological drought?

A. Drought can unfold from rainfall deficit through soil-moisture stress, storage decline, ecosystem damage, livelihood loss and health effects over different time scales, so one indicator cannot represent every stage. B. Ecological drought describes water shortage severe enough to impair ecosystem productivity or function; it must not be silently substituted for socio-economic drought. C. India's Manual for Drought Management uses rainfall, vegetative indices, crop-sowing progression, soil moisture and hydrological indices in a graded assessment rather than one universal crop-loss percentage. D. Meteorological drought concerns prolonged inadequate or poorly distributed precipitation relative to the relevant climatology; it is a hazard indicator, not a complete account of impacts.

Answer: D. Explanation: Meteorological drought concerns prolonged inadequate or poorly distributed precipitation relative to the relevant climatology; it is a hazard indicator, not a complete account of impacts. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q9. Which statement correctly identifies Agricultural drought?

A. Agricultural drought arises when soil moisture and crop-water availability become inadequate for crops, so rainfall information must be read with sowing, vegetation and soil-moisture evidence. B. Ecological drought describes water shortage severe enough to impair ecosystem productivity or function; it must not be silently substituted for socio-economic drought. C. Socio-economic drought occurs when water shortage disrupts the supply of goods, services, livelihoods or essential uses; it expresses the interaction of physical scarcity with demand and unequal access. D. Hydrological drought concerns below-normal availability in rivers, reservoirs, lakes or aquifers and may lag behind the rainfall deficit that initiated it.

Answer: A. Explanation: Agricultural drought arises when soil moisture and crop-water availability become inadequate for crops, so rainfall information must be read with sowing, vegetation and soil-moisture evidence. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q10. Which option preserves the risk or institutional boundary of Agricultural drought?

A. Drought can unfold from rainfall deficit through soil-moisture stress, storage decline, ecosystem damage, livelihood loss and health effects over different time scales, so one indicator cannot represent every stage. B. Agricultural drought arises when soil moisture and crop-water availability become inadequate for crops, so rainfall information must be read with sowing, vegetation and soil-moisture evidence. C. Ecological drought describes water shortage severe enough to impair ecosystem productivity or function; it must not be silently substituted for socio-economic drought. D. Socio-economic drought occurs when water shortage disrupts the supply of goods, services, livelihoods or essential uses; it expresses the interaction of physical scarcity with demand and unequal access.

Answer: B. Explanation: Agricultural drought arises when soil moisture and crop-water availability become inadequate for crops, so rainfall information must be read with sowing, vegetation and soil-moisture evidence. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q11. Which statement uses Agricultural drought without changing its hazard, mandate or status?

A. India's Manual for Drought Management uses rainfall, vegetative indices, crop-sowing progression, soil moisture and hydrological indices in a graded assessment rather than one universal crop-loss percentage. B. Ecological drought describes water shortage severe enough to impair ecosystem productivity or function; it must not be silently substituted for socio-economic drought. C. Agricultural drought arises when soil moisture and crop-water availability become inadequate for crops, so rainfall information must be read with sowing, vegetation and soil-moisture evidence. D. Drought can unfold from rainfall deficit through soil-moisture stress, storage decline, ecosystem damage, livelihood loss and health effects over different time scales, so one indicator cannot represent every stage.

Answer: C. Explanation: Agricultural drought arises when soil moisture and crop-water availability become inadequate for crops, so rainfall information must be read with sowing, vegetation and soil-moisture evidence. The other options change the hazard or risk category, institutional owner, legal instrument, warning

stage, evidence rung or implementation status.

Q12. Which option avoids the standard UPSC close-option trap about Agricultural drought?

A. India's Manual for Drought Management uses rainfall, vegetative indices, crop-sowing progression, soil moisture and hydrological indices in a graded assessment rather than one universal crop-loss percentage. B. A heat wave is a meteorological hazard identified through IMD's station and departure or actual-temperature criteria; it is not synonymous with every hot day or every episode of heat illness. C. Drought can unfold from rainfall deficit through soil-moisture stress, storage decline, ecosystem damage, livelihood loss and health effects over different time scales, so one indicator cannot represent every stage. D. Agricultural drought arises when soil moisture and crop-water availability become inadequate for crops, so rainfall information must be read with sowing, vegetation and soil-moisture evidence.

Answer: D. Explanation: Agricultural drought arises when soil moisture and crop-water availability become inadequate for crops, so rainfall information must be read with sowing, vegetation and soil-moisture evidence. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q13. Which statement correctly identifies Hydrological drought?

A. Hydrological drought concerns below-normal availability in rivers, reservoirs, lakes or aquifers and may lag behind the rainfall deficit that initiated it. B. Socio-economic drought occurs when water shortage disrupts the supply of goods, services, livelihoods or essential uses; it expresses the interaction of physical scarcity with demand and unequal access. C. Ecological drought describes water shortage severe enough to impair ecosystem productivity or function; it must not be silently substituted for socio-economic drought. D. Drought can unfold from rainfall deficit through soil-moisture stress, storage decline, ecosystem damage, livelihood loss and health effects over different time scales, so one indicator cannot represent every stage.

Answer: A. Explanation: Hydrological drought concerns below-normal availability in rivers, reservoirs, lakes or aquifers and may lag behind the rainfall deficit that initiated it. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q14. Which option preserves the risk or institutional boundary of Hydrological drought?

A. India's Manual for Drought Management uses rainfall, vegetative indices, crop-sowing progression, soil moisture and hydrological indices in a graded assessment rather than one universal crop-loss percentage. B. Hydrological drought concerns below-normal availability in rivers, reservoirs, lakes or aquifers and may lag behind the rainfall deficit that initiated it. C. Drought can unfold from rainfall deficit through soil-moisture stress, storage decline, ecosystem damage, livelihood loss and health effects over different time scales, so one indicator cannot represent every stage. D. Ecological drought describes water shortage severe enough to impair ecosystem productivity or function; it must not be silently substituted for socio-economic drought.

Answer: B. Explanation: Hydrological drought concerns below-normal availability in rivers, reservoirs, lakes or aquifers and may lag behind the rainfall deficit that initiated it. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q15. Which statement uses Hydrological drought without changing its hazard, mandate or status?

A. India's Manual for Drought Management uses rainfall, vegetative indices, crop-sowing progression, soil moisture and hydrological indices in a graded assessment rather than one universal crop-loss percentage. B. A heat wave is a meteorological hazard identified through IMD's station and departure or actual-temperature criteria; it is not synonymous with every hot day or every episode of heat illness. C. Hydrological drought concerns below-normal availability in rivers, reservoirs, lakes or aquifers and may lag behind the rainfall deficit that initiated it. D. Drought can unfold from rainfall deficit through soil-moisture stress, storage decline, ecosystem damage, livelihood loss and health effects over different time scales, so one indicator cannot represent every stage.

Answer: C. Explanation: Hydrological drought concerns below-normal availability in rivers, reservoirs, lakes or aquifers and may lag behind the rainfall deficit that initiated it. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q16. Which option avoids the standard UPSC close-option trap about Hydrological drought?

A. A heat wave is a meteorological hazard identified through IMD's station and departure or actual-temperature criteria; it is not synonymous with every hot day or every episode of heat illness. B. India's Manual for Drought Management uses rainfall, vegetative indices, crop-sowing progression, soil moisture and hydrological indices in a graded assessment rather than one universal crop-loss percentage. C. Heat stress is the physiological burden produced by temperature, humidity, radiation, wind, exertion, clothing, health and access to rest, water or cooling; exposure and vulnerability therefore mediate the hazard. D. Hydrological drought concerns below-normal availability in rivers, reservoirs, lakes or aquifers and may lag behind the rainfall deficit that initiated it.

Answer: D. Explanation: Hydrological drought concerns below-normal availability in rivers, reservoirs, lakes or aquifers and may lag behind the rainfall deficit that initiated it. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q17. Which statement correctly identifies Socio-economic drought?

A. Socio-economic drought occurs when water shortage disrupts the supply of goods, services, livelihoods or essential uses; it expresses the interaction of physical scarcity with demand and unequal access. B. India's Manual for Drought Management uses rainfall, vegetative indices, crop-sowing progression, soil moisture and hydrological indices in a graded assessment rather than one universal crop-loss percentage. C. Drought can unfold from rainfall deficit through soil-moisture stress, storage decline, ecosystem damage, livelihood loss and health effects over different time scales, so one indicator cannot represent every stage. D. Ecological drought describes water shortage severe enough to impair ecosystem productivity or function; it must not be silently substituted for socio-economic drought.

Answer: A. Explanation: Socio-economic drought occurs when water shortage disrupts the supply of goods, services, livelihoods or essential uses; it expresses the interaction of physical scarcity with demand and unequal access. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q18. Which option preserves the risk or institutional boundary of Socio-economic drought?

A. India's Manual for Drought Management uses rainfall, vegetative indices, crop-sowing progression, soil moisture and hydrological indices in a graded assessment rather than one universal crop-loss percentage. B. Socio-economic drought occurs when water shortage disrupts the supply of goods, services, livelihoods or essential uses; it expresses the interaction of physical scarcity with demand and unequal access. C. Drought can unfold from rainfall deficit through soil-moisture stress, storage decline, ecosystem damage, livelihood loss and health effects over different time scales, so one indicator cannot represent every stage. D. A heat wave is a meteorological hazard identified through IMD's station and departure or actual-temperature criteria; it is not synonymous with every hot day or every episode of heat illness.

Answer: B. Explanation: Socio-economic drought occurs when water shortage disrupts the supply of goods, services, livelihoods or essential uses; it expresses the interaction of physical scarcity with demand and unequal access. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q19. Which statement uses Socio-economic drought without changing its hazard, mandate or status?

A. India's Manual for Drought Management uses rainfall, vegetative indices, crop-sowing progression, soil moisture and hydrological indices in a graded assessment rather than one universal crop-loss percentage. B. A heat wave is a meteorological hazard identified through IMD's station and departure or actual-temperature criteria; it is not synonymous with every hot day or every episode of heat illness. C. Socio-economic drought occurs when water shortage disrupts the supply of goods, services, livelihoods or essential uses; it expresses the interaction of physical scarcity with demand and unequal access. D. Heat stress is the physiological burden produced by temperature, humidity, radiation, wind, exertion, clothing, health and access to rest, water or cooling; exposure and vulnerability therefore mediate the hazard.

Answer: C. Explanation: Socio-economic drought occurs when water shortage disrupts the supply of goods, services, livelihoods or essential uses; it expresses the interaction of physical scarcity with demand and unequal access. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q20. Which option avoids the standard UPSC close-option trap about Socio-economic drought?

A. Wet-bulb temperature combines heat and humidity by representing evaporative-cooling potential; it is not an IMD heat-wave declaration threshold, and dangerous stress can occur below a theoretical upper-limit benchmark. B. A heat wave is a meteorological hazard identified through IMD's station and departure or actual-temperature criteria; it is not synonymous with every hot day or every episode of heat illness. C. Heat stress is the physiological burden produced by temperature, humidity, radiation, wind, exertion, clothing, health and access to rest, water or cooling; exposure and vulnerability therefore mediate the hazard. D. Socio-economic drought occurs when water shortage disrupts the supply of goods, services, livelihoods or essential uses; it expresses the interaction of physical scarcity with demand and unequal access.

Answer: D. Explanation: Socio-economic drought occurs when water shortage disrupts the supply of goods, services, livelihoods or essential uses; it expresses the interaction of physical scarcity with demand and unequal access. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q21. Which statement correctly identifies Ecological drought?

A. Ecological drought describes water shortage severe enough to impair ecosystem productivity or function; it must not be silently substituted for socio-economic drought. B. A heat wave is a meteorological hazard identified through IMD's station and departure or actual-temperature criteria; it is not synonymous with every hot day or every episode of heat illness. C. Drought can unfold from rainfall deficit through soil-moisture stress, storage decline, ecosystem damage, livelihood loss and health effects over different time scales, so one indicator cannot represent every stage. D. India's Manual for Drought Management uses rainfall, vegetative indices, crop-sowing progression, soil moisture and hydrological indices in a graded assessment rather than one universal crop-loss percentage.

Answer: A. Explanation: Ecological drought describes water shortage severe enough to impair ecosystem productivity or function; it must not be silently substituted for socio-economic drought. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q22. Which option preserves the risk or institutional boundary of Ecological drought?

A. Heat stress is the physiological burden produced by temperature, humidity, radiation, wind, exertion, clothing, health and access to rest, water or cooling; exposure and vulnerability therefore mediate the hazard. B. Ecological drought describes water shortage severe enough to impair ecosystem productivity or function; it must not be silently substituted for socio-economic drought. C. India's Manual for Drought Management uses rainfall, vegetative indices, crop-sowing progression, soil moisture and hydrological indices in a graded assessment rather than one universal crop-loss percentage. D. A heat wave is a meteorological hazard identified through IMD's station and departure or actual-temperature criteria; it is not synonymous with every hot day or every episode of heat illness.

Answer: B. Explanation: Ecological drought describes water shortage severe enough to impair ecosystem productivity or function; it must not be silently substituted for socio-economic drought. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q23. Which statement uses Ecological drought without changing its hazard, mandate or status?

A. Heat stress is the physiological burden produced by temperature, humidity, radiation, wind, exertion, clothing, health and access to rest, water or cooling; exposure and vulnerability therefore mediate the hazard. B. Wet-bulb temperature combines heat and humidity by representing evaporative-cooling potential; it is not an IMD heat-wave declaration threshold, and dangerous stress can occur below a theoretical upper-limit benchmark. C. Ecological drought describes water shortage severe enough to impair ecosystem productivity or function; it must not be silently substituted for socio-economic drought. D. A heat wave is a meteorological hazard identified through IMD's station and departure or actual-temperature criteria; it is not synonymous with every hot day or every episode of heat illness.

Answer: C. Explanation: Ecological drought describes water shortage severe enough to impair ecosystem productivity or function; it must not be silently substituted for socio-economic drought. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q24. Which option avoids the standard UPSC close-option trap about Ecological drought?

A. Heat stress is the physiological burden produced by temperature, humidity, radiation, wind, exertion, clothing, health and access to rest, water or cooling; exposure and vulnerability therefore mediate the hazard. B. Outdoor and informal workers, older persons, children, people with illness, poorly housed households and those lacking dependable water or cooling may face different risk under the same forecast temperature. C. Wet-bulb temperature combines heat and humidity by representing evaporative-cooling potential; it is not an IMD heat-wave declaration threshold, and dangerous stress can occur below a theoretical upper-limit benchmark. D. Ecological drought describes water shortage severe enough to impair ecosystem productivity or function; it must not be silently substituted for socio-economic drought.

Answer: D. Explanation: Ecological drought describes water shortage severe enough to impair ecosystem productivity or function; it must not be silently substituted for socio-economic drought. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q25. Which statement correctly identifies Slow-onset cascade?

A. Drought can unfold from rainfall deficit through soil-moisture stress, storage decline, ecosystem damage, livelihood loss and health effects over different time scales, so one indicator cannot represent every stage. B. India's Manual for Drought Management uses rainfall, vegetative indices, crop-sowing progression, soil moisture and hydrological indices in a graded assessment rather than one universal crop-loss percentage. C. A heat wave is a meteorological hazard identified through IMD's station and departure or actual-temperature criteria; it is not synonymous with every hot day or every episode of heat illness. D. Heat stress is the physiological burden produced by temperature, humidity, radiation, wind, exertion, clothing, health and access to rest, water or cooling; exposure and vulnerability therefore mediate the hazard.

Answer: A. Explanation: Drought can unfold from rainfall deficit through soil-moisture stress, storage decline, ecosystem damage, livelihood loss and health effects over different time scales, so one indicator cannot represent every stage. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q26. Which option preserves the risk or institutional boundary of Slow-onset cascade?

A. Heat stress is the physiological burden produced by temperature, humidity, radiation, wind, exertion, clothing, health and access to rest, water or cooling; exposure and vulnerability therefore mediate the hazard. B. Drought can unfold from rainfall deficit through soil-moisture stress, storage decline, ecosystem damage, livelihood loss and health effects over different time scales, so one indicator cannot represent every stage. C. A heat wave is a meteorological hazard identified through IMD's station and departure or actual-temperature criteria; it is not synonymous with every hot day or every episode of heat illness. D. Wet-bulb temperature combines heat and humidity by representing evaporative-cooling potential; it is not an IMD heat-wave declaration threshold, and dangerous stress can occur below a theoretical upper-limit benchmark.

Answer: B. Explanation: Drought can unfold from rainfall deficit through soil-moisture stress, storage decline, ecosystem damage, livelihood loss and health effects over different time scales, so one indicator cannot represent every stage. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q27. Which statement uses Slow-onset cascade without changing its hazard, mandate or status?

A. Wet-bulb temperature combines heat and humidity by representing evaporative-cooling potential; it is not an IMD heat-wave declaration threshold, and dangerous stress can occur below a theoretical upper-limit benchmark. B. Outdoor and informal workers, older persons, children, people with illness, poorly housed households and those lacking dependable water or cooling may face different risk under the same forecast temperature. C. Drought can unfold from rainfall deficit through soil-moisture stress, storage decline, ecosystem damage, livelihood loss and health effects over different time scales, so one indicator cannot represent every stage. D. Heat stress is the physiological burden produced by temperature, humidity, radiation, wind, exertion, clothing, health and access to rest, water or cooling; exposure and vulnerability therefore mediate the hazard.

Answer: C. Explanation: Drought can unfold from rainfall deficit through soil-moisture stress, storage decline, ecosystem damage, livelihood loss and health effects over different time scales, so one indicator cannot represent every stage. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q28. Which option avoids the standard UPSC close-option trap about Slow-onset cascade?

A. Wet-bulb temperature combines heat and humidity by representing evaporative-cooling potential; it is not an IMD heat-wave declaration threshold, and dangerous stress can occur below a theoretical upper-limit benchmark. B. A Heat Action Plan links seasonal preparedness, warnings, health-system readiness, public communication, water and cooling access, work or school timing and longer-term urban measures through locally assigned responsibilities. C. Outdoor and informal workers, older persons, children, people with illness, poorly housed households and those lacking dependable water or cooling may face different risk under the same forecast temperature. D. Drought can unfold from rainfall deficit through soil-moisture stress, storage decline, ecosystem damage, livelihood loss and health effects over different time scales, so one indicator cannot represent every stage.

Answer: D. Explanation: Drought can unfold from rainfall deficit through soil-moisture stress, storage decline, ecosystem damage, livelihood loss and health effects over different time scales, so one indicator cannot represent every stage. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q29. Which statement correctly identifies Multi-indicator declaration?

A. India's Manual for Drought Management uses rainfall, vegetative indices, crop-sowing progression, soil moisture and hydrological indices in a graded assessment rather than one universal crop-loss percentage. B. Wet-bulb temperature combines heat and humidity by representing evaporative-cooling potential; it is not an IMD heat-wave declaration threshold, and dangerous stress can occur below a theoretical upper-limit benchmark. C. A heat wave is a meteorological hazard identified through IMD's station and departure or actual-temperature criteria; it is not synonymous with every hot day or every episode of heat illness. D. Heat stress is the physiological burden produced by temperature, humidity, radiation, wind, exertion, clothing, health and access to rest, water or cooling; exposure and vulnerability therefore mediate the hazard.

Answer: A. Explanation: India's Manual for Drought Management uses rainfall, vegetative indices, crop-sowing progression, soil moisture and hydrological indices in a graded assessment rather than one universal crop-loss percentage. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q30. Which option preserves the risk or institutional boundary of Multi-indicator declaration?

A. Wet-bulb temperature combines heat and humidity by representing evaporative-cooling potential; it is not an IMD heat-wave declaration threshold, and dangerous stress can occur below a theoretical upper-limit benchmark. B. India's Manual for Drought Management uses rainfall, vegetative indices, crop-sowing progression, soil moisture and hydrological indices in a graded assessment rather than one universal crop-loss percentage. C. Outdoor and informal workers, older persons, children, people with illness, poorly housed households and those lacking dependable water or cooling may face different risk under the same forecast temperature. D. Heat stress is the physiological burden produced by temperature, humidity, radiation, wind, exertion, clothing, health and access to rest, water or cooling; exposure and vulnerability therefore mediate the hazard.

Answer: B. Explanation: India's Manual for Drought Management uses rainfall, vegetative indices, crop-sowing progression, soil moisture and hydrological indices in a graded assessment rather than one universal crop-loss percentage. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q31. Which statement uses Multi-indicator declaration without changing its hazard, mandate or status?

A. Wet-bulb temperature combines heat and humidity by representing evaporative-cooling potential; it is not an IMD heat-wave declaration threshold, and dangerous stress can occur below a theoretical upper-limit benchmark. B. A Heat Action Plan links seasonal preparedness, warnings, health-system readiness, public communication, water and cooling access, work or school timing and longer-term urban measures through locally assigned responsibilities. C. India's Manual for Drought Management uses rainfall, vegetative indices, crop-sowing progression, soil moisture and hydrological indices in a graded assessment rather than one universal crop-loss percentage. D. Outdoor and informal workers, older persons, children, people with illness, poorly housed households and those lacking dependable water or cooling may face different risk under the same forecast temperature.

Answer: C. Explanation: India's Manual for Drought Management uses rainfall, vegetative indices, crop-sowing progression, soil moisture and hydrological indices in a graded assessment rather than one universal crop-loss percentage. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q32. Which option avoids the standard UPSC close-option trap about Multi-indicator declaration?

A. Drought management requires demand management, drinking-water security, groundwater and surface-water monitoring, watershed or recharge measures, storage governance and protection against shifting scarcity to weaker users. B. A Heat Action Plan links seasonal preparedness, warnings, health-system readiness, public communication, water and cooling access, work or school timing and longer-term urban measures through locally assigned responsibilities. C. Outdoor and informal workers, older persons, children, people with illness, poorly housed households and those lacking dependable water or cooling may face different risk under the same forecast temperature. D. India's Manual for Drought Management uses rainfall, vegetative indices, crop-sowing progression, soil moisture and hydrological indices in a graded assessment rather than one universal crop-loss percentage.

Answer: D. Explanation: India's Manual for Drought Management uses rainfall, vegetative indices, crop-sowing progression, soil moisture and hydrological indices in a graded assessment rather than one universal crop-loss percentage. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q33. Which statement correctly identifies Heat-wave hazard?

A. A heat wave is a meteorological hazard identified through IMD's station and departure or actual-temperature criteria; it is not synonymous with every hot day or every episode of heat illness. B. Heat stress is the physiological burden produced by temperature, humidity, radiation, wind, exertion, clothing, health and access to rest, water or cooling; exposure and vulnerability therefore mediate the hazard. C. Outdoor and informal workers, older persons, children, people with illness, poorly housed households and those lacking dependable water or cooling may face different risk under the same forecast temperature. D. Wet-bulb temperature combines heat and humidity by representing evaporative-cooling potential; it is not an IMD heat-wave declaration threshold, and dangerous stress can occur below a theoretical upper-limit benchmark.

Answer: A. Explanation: A heat wave is a meteorological hazard identified through IMD's station and departure or actual-temperature criteria; it is not synonymous with every hot day or every episode of heat illness. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q34. Which option preserves the risk or institutional boundary of Heat-wave hazard?

A. Outdoor and informal workers, older persons, children, people with illness, poorly housed households and those lacking dependable water or cooling may face different risk under the same forecast temperature. B. A heat wave is a meteorological hazard identified through IMD's station and departure or actual-temperature criteria; it is not synonymous with every hot day or every episode of heat illness. C. Wet-bulb temperature combines heat and humidity by representing evaporative-cooling potential; it is not an IMD heat-wave declaration threshold, and dangerous stress can occur below a theoretical upper-limit benchmark. D. A Heat Action Plan links seasonal preparedness, warnings, health-system readiness, public communication, water and cooling access, work or school timing and longer-term urban measures through locally assigned responsibilities.

Answer: B. Explanation: A heat wave is a meteorological hazard identified through IMD's station and departure or actual-temperature criteria; it is not synonymous with every hot day or every episode of heat illness. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q35. Which statement uses Heat-wave hazard without changing its hazard, mandate or status?

A. Drought management requires demand management, drinking-water security, groundwater and surface-water monitoring, watershed or recharge measures, storage governance and protection against shifting scarcity to weaker users. B. A Heat Action Plan links seasonal preparedness, warnings, health-system readiness, public communication, water and cooling access, work or school timing and longer-term urban measures through locally assigned responsibilities. C. A heat wave is a meteorological hazard identified through IMD's station and departure or actual-temperature criteria; it is not synonymous with every hot day or every episode of heat illness. D. Outdoor and informal workers, older persons, children, people with illness, poorly housed households and those lacking dependable water or cooling may face different risk under the same forecast temperature.

Answer: C. Explanation: A heat wave is a meteorological hazard identified through IMD's station and departure or actual-temperature criteria; it is not synonymous with every hot day or every episode of heat illness. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q36. Which option avoids the standard UPSC close-option trap about Heat-wave hazard?

A. Crop choice, contingency planning, soil-moisture conservation, livestock support, insurance or social protection and alternative livelihoods address agricultural and socioeconomic drought without treating relief as adaptation. B. Drought management requires demand management, drinking-water security, groundwater and surface-water monitoring, watershed or recharge measures, storage governance and protection against shifting scarcity to weaker users. C. A Heat Action Plan links seasonal preparedness, warnings, health-system readiness, public communication, water and cooling access, work or school timing and longer-term urban measures through locally assigned responsibilities. D. A heat wave is a meteorological hazard identified through IMD's station and departure or actual-temperature criteria; it is not synonymous with every hot day or every episode of heat illness.

Answer: D. Explanation: A heat wave is a meteorological hazard identified through IMD's station and departure or actual-temperature criteria; it is not synonymous with every hot day or every episode of heat illness. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q37. Which statement correctly identifies Heat stress?

A. Heat stress is the physiological burden produced by temperature, humidity, radiation, wind, exertion, clothing, health and access to rest, water or cooling; exposure and vulnerability therefore mediate the hazard. B. Outdoor and informal workers, older persons, children, people with illness, poorly housed households and those lacking dependable water or cooling may face different risk under the same forecast temperature. C. Wet-bulb temperature combines heat and humidity by representing evaporative-cooling potential; it is not an IMD heat-wave declaration threshold, and dangerous stress can occur below a theoretical upper-limit benchmark. D. A Heat Action Plan links seasonal preparedness, warnings, health-system readiness, public communication, water and cooling access, work or school timing and longer-term urban measures through locally assigned responsibilities.

Answer: A. Explanation: Heat stress is the physiological burden produced by temperature, humidity, radiation, wind, exertion, clothing, health and access to rest, water or cooling; exposure and vulnerability therefore mediate the hazard. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q38. Which option preserves the risk or institutional boundary of Heat stress?

A. Outdoor and informal workers, older persons, children, people with illness, poorly housed households and those lacking dependable water or cooling may face different risk under the same forecast temperature. B. Heat stress is the physiological burden produced by temperature, humidity, radiation, wind, exertion, clothing, health and access to rest, water or cooling; exposure and vulnerability therefore mediate the hazard. C. Drought management requires demand management, drinking-water security, groundwater and surface-water monitoring, watershed or recharge measures, storage governance and protection against shifting scarcity to weaker users. D. A Heat Action Plan links seasonal preparedness, warnings, health-system readiness, public communication, water and cooling access, work or school timing and longer-term urban measures through locally assigned responsibilities.

Answer: B. Explanation: Heat stress is the physiological burden produced by temperature, humidity, radiation, wind, exertion, clothing, health and access to rest, water or cooling; exposure and vulnerability therefore mediate the hazard. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q39. Which statement uses Heat stress without changing its hazard, mandate or status?

A. Drought management requires demand management, drinking-water security, groundwater and surface-water monitoring, watershed or recharge measures, storage governance and protection against shifting scarcity to weaker users. B. A Heat Action Plan links seasonal preparedness, warnings, health-system readiness, public communication, water and cooling access, work or school timing and longer-term urban measures through locally assigned responsibilities. C. Heat stress is the physiological burden produced by temperature, humidity, radiation, wind, exertion, clothing, health and access to rest, water or cooling; exposure and vulnerability therefore mediate the hazard. D. Crop choice, contingency planning, soil-moisture conservation, livestock support, insurance or social protection and alternative livelihoods address agricultural and socioeconomic drought without treating relief as adaptation.

Answer: C. Explanation: Heat stress is the physiological burden produced by temperature, humidity, radiation, wind, exertion, clothing, health and access to rest, water or cooling; exposure and vulnerability therefore mediate the hazard. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q40. Which option avoids the standard UPSC close-option trap about Heat stress?

A. Cool roofs, shade, ventilation, trees and water bodies can reduce selected exposures, but benefits depend on design, maintenance, local climate and access; a cooling intervention can transfer costs or water demand. B. Crop choice, contingency planning, soil-moisture conservation, livestock support, insurance or social protection and alternative livelihoods address agricultural and socioeconomic drought without treating relief as adaptation. C. Drought management requires demand management, drinking-water security, groundwater and surface-water monitoring, watershed or recharge measures, storage governance and protection against shifting scarcity to weaker users. D. Heat stress is the physiological burden produced by temperature, humidity, radiation, wind, exertion, clothing, health and access to rest, water or cooling; exposure and vulnerability therefore mediate the hazard.

Answer: D. Explanation: Heat stress is the physiological burden produced by temperature, humidity, radiation, wind, exertion, clothing, health and access to rest, water or cooling; exposure and vulnerability therefore mediate the hazard. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q41. Which statement correctly identifies Wet-bulb boundary?

A. Wet-bulb temperature combines heat and humidity by representing evaporative-cooling potential; it is not an IMD heat-wave declaration threshold, and dangerous stress can occur below a theoretical upper-limit benchmark. B. A Heat Action Plan links seasonal preparedness, warnings, health-system readiness, public communication, water and cooling access, work or school timing and longer-term urban measures through locally assigned responsibilities. C. Drought management requires demand management, drinking-water security, groundwater and surface-water monitoring, watershed or recharge measures, storage governance and protection against shifting scarcity to weaker users. D. Outdoor and informal workers, older persons, children, people with illness, poorly housed households and those lacking dependable water or cooling may face different risk under the same forecast temperature.

Answer: A. Explanation: Wet-bulb temperature combines heat and humidity by representing evaporative-cooling potential; it is not an IMD heat-wave declaration threshold, and dangerous stress can occur below a theoretical upper-limit benchmark. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q42. Which option preserves the risk or institutional boundary of Wet-bulb boundary?

A. Crop choice, contingency planning, soil-moisture conservation, livestock support, insurance or social protection and alternative livelihoods address agricultural and socioeconomic drought without treating relief as adaptation. B. Wet-bulb temperature combines heat and humidity by representing evaporative-cooling potential; it is not an IMD heat-wave declaration threshold, and dangerous stress can occur below a theoretical upper-limit benchmark. C. Drought management requires demand management, drinking-water security, groundwater and surface-water monitoring, watershed or recharge measures, storage governance and protection against shifting scarcity to weaker users. D. A Heat Action Plan links seasonal preparedness, warnings, health-system readiness, public communication, water and cooling access, work or school timing and longer-term urban measures through locally assigned responsibilities.

Answer: B. Explanation: Wet-bulb temperature combines heat and humidity by representing evaporative-cooling potential; it is not an IMD heat-wave declaration threshold, and dangerous stress can occur below a theoretical upper-limit benchmark. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q43. Which statement uses Wet-bulb boundary without changing its hazard, mandate or status?

A. Cool roofs, shade, ventilation, trees and water bodies can reduce selected exposures, but benefits depend on design, maintenance, local climate and access; a cooling intervention can transfer costs or water demand. B. Drought management requires demand management, drinking-water security, groundwater and surface-water monitoring, watershed or recharge measures, storage governance and protection against shifting scarcity to weaker users. C. Wet-bulb temperature combines heat and humidity by representing evaporative-cooling potential; it is not an IMD heat-wave declaration threshold, and dangerous stress can occur below a theoretical upper-limit benchmark. D. Crop choice, contingency planning, soil-moisture conservation, livestock support, insurance or social protection and alternative livelihoods address agricultural and socioeconomic drought without treating relief as adaptation.

Answer: C. Explanation: Wet-bulb temperature combines heat and humidity by representing evaporative-cooling potential; it is not an IMD heat-wave declaration threshold, and dangerous stress can occur below a theoretical upper-limit benchmark. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q44. Which option avoids the standard UPSC close-option trap about Wet-bulb boundary?

A. Cool roofs, shade, ventilation, trees and water bodies can reduce selected exposures, but benefits depend on design, maintenance, local climate and access; a cooling intervention can transfer costs or water demand. B. Crop choice, contingency planning, soil-moisture conservation, livestock support, insurance or social protection and alternative livelihoods address agricultural and socioeconomic drought without treating relief as adaptation. C. Risk-sensitive work-rest scheduling, drinking water, shaded rest, worker communication, clinical readiness and surveillance are distinct from the meteorological forecast and require responsible employers and public agencies. D. Wet-bulb temperature combines heat and humidity by representing evaporative-cooling potential; it is not an IMD heat-wave declaration threshold, and dangerous stress can occur below a theoretical upper-limit benchmark.

Answer: D. Explanation: Wet-bulb temperature combines heat and humidity by representing evaporative-cooling potential; it is not an IMD heat-wave declaration threshold, and dangerous stress can occur below a theoretical upper-limit benchmark. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q45. Which statement correctly identifies Exposure inequality?

A. Outdoor and informal workers, older persons, children, people with illness, poorly housed households and those lacking dependable water or cooling may face different risk under the same forecast temperature. B. Drought management requires demand management, drinking-water security, groundwater and surface-water monitoring, watershed or recharge measures, storage governance and protection against shifting scarcity to weaker users. C. A Heat Action Plan links seasonal preparedness, warnings, health-system readiness, public communication, water and cooling access, work or school timing and longer-term urban measures through locally assigned responsibilities. D. Crop choice, contingency planning, soil-moisture conservation, livestock support, insurance or social protection and alternative livelihoods address agricultural and socioeconomic drought without treating relief as adaptation.

Answer: A. Explanation: Outdoor and informal workers, older persons, children, people with illness, poorly housed households and those lacking dependable water or cooling may face different risk under the same forecast temperature. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q46. Which option preserves the risk or institutional boundary of Exposure inequality?

A. Crop choice, contingency planning, soil-moisture conservation, livestock support, insurance or social protection and alternative livelihoods address agricultural and socioeconomic drought without treating relief as adaptation. B. Outdoor and informal workers, older persons, children, people with illness, poorly housed households and those lacking dependable water or cooling may face different risk under the same forecast temperature. C. Drought management requires demand management, drinking-water security, groundwater and surface-water monitoring, watershed or recharge measures, storage governance and protection against shifting scarcity to weaker users. D. Cool roofs, shade, ventilation, trees and water bodies can reduce selected exposures, but benefits depend on design, maintenance, local climate and access; a cooling intervention can transfer costs or water demand.

Answer: B. Explanation: Outdoor and informal workers, older persons, children, people with illness, poorly housed households and those lacking dependable water or cooling may face different risk under the same forecast temperature. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q47. Which statement uses Exposure inequality without changing its hazard, mandate or status?

A. Cool roofs, shade, ventilation, trees and water bodies can reduce selected exposures, but benefits depend on design, maintenance, local climate and access; a cooling intervention can transfer costs or water demand. B. Crop choice, contingency planning, soil-moisture conservation, livestock support, insurance or social protection and alternative livelihoods address agricultural and socioeconomic drought without treating relief as adaptation. C. Outdoor and informal workers, older persons, children, people with illness, poorly housed households and those lacking dependable water or cooling may face different risk under the same forecast temperature. D. Risk-sensitive work-rest scheduling, drinking water, shaded rest, worker communication, clinical readiness and surveillance are distinct from the meteorological forecast and require responsible employers and public agencies.

Answer: C. Explanation: Outdoor and informal workers, older persons, children, people with illness, poorly housed households and those lacking dependable water or cooling may face different risk under the same forecast temperature. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q48. Which option avoids the standard UPSC close-option trap about Exposure inequality?

A. Risk-sensitive work-rest scheduling, drinking water, shaded rest, worker communication, clinical readiness and surveillance are distinct from the meteorological forecast and require responsible employers and public agencies. B. Cool roofs, shade, ventilation, trees and water bodies can reduce selected exposures, but benefits depend on design, maintenance, local climate and access; a cooling intervention can transfer costs or water demand. C. Drought, water scarcity, heat, wildfire risk, crop stress, power demand and public-health pressure can interact, while humid heat or hot nights may intensify harm without changing the formal hazard label. D. Outdoor and informal workers, older persons, children, people with illness, poorly housed households and those lacking dependable water or cooling may face different risk under the same forecast temperature.

Answer: D. Explanation: Outdoor and informal workers, older persons, children, people with illness, poorly housed households and those lacking dependable water or cooling may face different risk under the same forecast temperature. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q49. Which statement correctly identifies Heat Action Plan?

A. A Heat Action Plan links seasonal preparedness, warnings, health-system readiness, public communication, water and cooling access, work or school timing and longer-term urban measures through locally assigned responsibilities. B. Crop choice, contingency planning, soil-moisture conservation, livestock support, insurance or social protection and alternative livelihoods address agricultural and socioeconomic drought without treating relief as adaptation. C. Drought management requires demand management, drinking-water security, groundwater and surface-water monitoring, watershed or recharge measures, storage governance and protection against shifting scarcity to weaker users. D. Cool roofs, shade, ventilation, trees and water bodies can reduce selected exposures, but benefits depend on design, maintenance, local climate and access; a cooling intervention can transfer costs or water demand.

Answer: A. Explanation: A Heat Action Plan links seasonal preparedness, warnings, health-system readiness, public communication, water and cooling access, work or school timing and longer-term urban measures through locally assigned responsibilities. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q50. Which option preserves the risk or institutional boundary of Heat Action Plan?

A. Cool roofs, shade, ventilation, trees and water bodies can reduce selected exposures, but benefits depend on design, maintenance, local climate and access; a cooling intervention can transfer costs or water demand. B. A Heat Action Plan links seasonal preparedness, warnings, health-system readiness, public communication, water and cooling access, work or school timing and longer-term urban measures through locally assigned responsibilities. C. Crop choice, contingency planning, soil-moisture conservation, livestock support, insurance or social protection and alternative livelihoods address agricultural and socioeconomic drought without treating relief as adaptation. D. Risk-sensitive work-rest scheduling, drinking water, shaded rest, worker communication, clinical readiness and surveillance are distinct from the meteorological forecast and require responsible employers and public agencies.

Answer: B. Explanation: A Heat Action Plan links seasonal preparedness, warnings, health-system readiness, public communication, water and cooling access, work or school timing and longer-term urban measures through locally assigned responsibilities. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q51. Which statement uses Heat Action Plan without changing its hazard, mandate or status?

A. Cool roofs, shade, ventilation, trees and water bodies can reduce selected exposures, but benefits depend on design, maintenance, local climate and access; a cooling intervention can transfer costs or water demand. B. Risk-sensitive work-rest scheduling, drinking water, shaded rest, worker communication, clinical readiness and surveillance are distinct from the meteorological forecast and require responsible employers and public agencies. C. A Heat Action Plan links seasonal preparedness, warnings, health-system readiness, public communication, water and cooling access, work or school timing and longer-term urban measures through locally assigned responsibilities. D. Drought, water scarcity, heat, wildfire risk, crop stress, power demand and public-health pressure can interact, while humid heat or hot nights may intensify harm without changing the formal hazard label.

Answer: C. Explanation: A Heat Action Plan links seasonal preparedness, warnings, health-system readiness, public communication, water and cooling access, work or school timing and longer-term urban measures through locally assigned responsibilities. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q52. Which option avoids the standard UPSC close-option trap about Heat Action Plan?

A. Drought, water scarcity, heat, wildfire risk, crop stress, power demand and public-health pressure can interact, while humid heat or hot nights may intensify harm without changing the formal hazard label. B. Risk-sensitive work-rest scheduling, drinking water, shaded rest, worker communication, clinical readiness and surveillance are distinct from the meteorological forecast and require responsible employers and public agencies. C. Forecast-triggered early action should precede severe impacts, but water-intensive cooling, indiscriminate groundwater extraction or unequal greening can increase future vulnerability and become maladaptation. D. A Heat Action Plan links seasonal preparedness, warnings, health-system readiness, public communication, water and cooling access, work or school timing and longer-term urban measures through locally assigned responsibilities.

Answer: D. Explanation: A Heat Action Plan links seasonal preparedness, warnings, health-system readiness, public communication, water and cooling access, work or school timing and longer-term urban measures through locally assigned responsibilities. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q53. Which statement correctly identifies Water-risk portfolio?

A. Drought management requires demand management, drinking-water security, groundwater and surface-water monitoring, watershed or recharge measures, storage governance and protection against shifting scarcity to weaker users. B. Cool roofs, shade, ventilation, trees and water bodies can reduce selected exposures, but benefits depend on design, maintenance, local climate and access; a cooling intervention can transfer costs or water demand. C. Crop choice, contingency planning, soil-moisture conservation, livestock support, insurance or social protection and alternative livelihoods address agricultural and socioeconomic drought without treating relief as adaptation. D. Risk-sensitive work-rest scheduling, drinking water, shaded rest, worker communication, clinical readiness and surveillance are distinct from the meteorological forecast and require responsible employers and public agencies.

Answer: A. Explanation: Drought management requires demand management, drinking-water security, groundwater and surface-water monitoring, watershed or recharge measures, storage governance and protection against shifting scarcity to weaker users. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q54. Which option preserves the risk or institutional boundary of Water-risk portfolio?

A. Risk-sensitive work-rest scheduling, drinking water, shaded rest, worker communication, clinical readiness and surveillance are distinct from the meteorological forecast and require responsible employers and public agencies. B. Drought management requires demand management, drinking-water security, groundwater and surface-water monitoring, watershed or recharge measures, storage governance and protection against shifting scarcity to weaker users. C. Drought, water scarcity, heat, wildfire risk, crop stress, power demand and public-health pressure can interact, while humid heat or hot nights may intensify harm without changing the formal hazard label. D. Cool roofs, shade, ventilation, trees and water bodies can reduce selected exposures, but benefits depend on design, maintenance, local climate and access; a cooling intervention can transfer costs or water demand.

Answer: B. Explanation: Drought management requires demand management, drinking-water security, groundwater and surface-water monitoring, watershed or recharge measures, storage governance and protection against shifting scarcity to weaker users. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q55. Which statement uses Water-risk portfolio without changing its hazard, mandate or status?

A. Risk-sensitive work-rest scheduling, drinking water, shaded rest, worker communication, clinical readiness and surveillance are distinct from the meteorological forecast and require responsible employers and public agencies. B. Forecast-triggered early action should precede severe impacts, but water-intensive cooling, indiscriminate groundwater extraction or unequal greening can increase future vulnerability and become maladaptation. C. Drought management requires demand management, drinking-water security, groundwater and surface-water monitoring, watershed or recharge measures, storage governance and protection against shifting scarcity to weaker users. D. Drought, water scarcity, heat, wildfire risk, crop stress, power demand and public-health pressure can interact, while humid heat or hot nights may intensify harm without changing the formal hazard label.

Answer: C. Explanation: Drought management requires demand management, drinking-water security, groundwater and surface-water monitoring, watershed or recharge measures, storage governance and protection against shifting scarcity to weaker users. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q56. Which option avoids the standard UPSC close-option trap about Water-risk portfolio?

A. Forecast-triggered early action should precede severe impacts, but water-intensive cooling, indiscriminate groundwater extraction or unequal greening can increase future vulnerability and become maladaptation. B. Drought, water scarcity, heat, wildfire risk, crop stress, power demand and public-health pressure can interact, while humid heat or hot nights may intensify harm without changing the formal hazard label. C. A drought declaration, forecast, Heat Action Plan, dashboard, water scheme or contingency plan proves a classification or input; timely reach, equitable access, livelihood continuity and reduced illness require separate evidence. D. Drought management requires demand management, drinking-water security, groundwater and surface-water monitoring, watershed or recharge measures, storage governance and protection against shifting scarcity to weaker users.

Answer: D. Explanation: Drought management requires demand management, drinking-water security, groundwater and surface-water monitoring, watershed or recharge measures, storage governance and protection against shifting scarcity to weaker users. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q57. Which statement correctly identifies Agriculture and livelihoods?

A. Crop choice, contingency planning, soil-moisture conservation, livestock support, insurance or social protection and alternative livelihoods address agricultural and socioeconomic drought without treating relief as adaptation. B. Risk-sensitive work-rest scheduling, drinking water, shaded rest, worker communication, clinical readiness and surveillance are distinct from the meteorological forecast and require responsible employers and public agencies. C. Drought, water scarcity, heat, wildfire risk, crop stress, power demand and public-health pressure can interact, while humid heat or hot nights may intensify harm without changing the formal hazard label. D. Cool roofs, shade, ventilation, trees and water bodies can reduce selected exposures, but benefits depend on design, maintenance, local climate and access; a cooling intervention can transfer costs or water demand.

Answer: A. Explanation: Crop choice, contingency planning, soil-moisture conservation, livestock support, insurance or social protection and alternative livelihoods address agricultural and socioeconomic drought without treating relief as adaptation. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q58. Which option preserves the risk or institutional boundary of Agriculture and livelihoods?

A. Forecast-triggered early action should precede severe impacts, but water-intensive cooling, indiscriminate groundwater extraction or unequal greening can increase future vulnerability and become maladaptation. B. Crop choice, contingency planning, soil-moisture conservation, livestock support, insurance or social protection and alternative livelihoods address agricultural and socioeconomic drought without treating relief as adaptation. C. Drought, water scarcity, heat, wildfire risk, crop stress, power demand and public-health pressure can interact, while humid heat or hot nights may intensify harm without changing the formal hazard label. D. Risk-sensitive work-rest scheduling, drinking water, shaded rest, worker communication, clinical readiness and surveillance are distinct from the meteorological forecast and require responsible employers and public agencies.

Answer: B. Explanation: Crop choice, contingency planning, soil-moisture conservation, livestock support, insurance or social protection and alternative livelihoods address agricultural and socioeconomic drought without treating relief as adaptation. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q59. Which statement uses Agriculture and livelihoods without changing its hazard, mandate or status?

A. A drought declaration, forecast, Heat Action Plan, dashboard, water scheme or contingency plan proves a classification or input; timely reach, equitable access, livelihood continuity and reduced illness require separate evidence. B. Drought, water scarcity, heat, wildfire risk, crop stress, power demand and public-health pressure can interact, while humid heat or hot nights may intensify harm without changing the formal hazard label. C. Crop choice, contingency planning, soil-moisture conservation, livestock support, insurance or social protection and alternative livelihoods address agricultural and socioeconomic drought without treating relief as adaptation. D. Forecast-triggered early action should precede severe impacts, but water-intensive cooling, indiscriminate groundwater extraction or unequal greening can increase future vulnerability and become maladaptation.

Answer: C. Explanation: Crop choice, contingency planning, soil-moisture conservation, livestock support, insurance or social protection and alternative livelihoods address agricultural and socioeconomic drought without treating relief as adaptation. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q60. Which option avoids the standard UPSC close-option trap about Agriculture and livelihoods?

A. Forecast-triggered early action should precede severe impacts, but water-intensive cooling, indiscriminate groundwater extraction or unequal greening can increase future vulnerability and become maladaptation. B. Aridity is a long-term climatic condition of low moisture availability, whereas drought is a temporary or episodic deficit relative to the expected local water regime. C. A drought declaration, forecast, Heat Action Plan, dashboard, water scheme or contingency plan proves a classification or input; timely reach, equitable access, livelihood continuity and reduced illness require separate evidence. D. Crop choice, contingency planning, soil-moisture conservation, livestock support, insurance or social protection and alternative livelihoods address agricultural and socioeconomic drought without treating relief as adaptation.

Answer: D. Explanation: Crop choice, contingency planning, soil-moisture conservation, livestock support, insurance or social protection and alternative livelihoods address agricultural and socioeconomic drought without treating relief as adaptation. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q61. Which statement correctly identifies Urban heat measures?

A. Cool roofs, shade, ventilation, trees and water bodies can reduce selected exposures, but benefits depend on design, maintenance, local climate and access; a cooling intervention can transfer costs or water demand. B. Risk-sensitive work-rest scheduling, drinking water, shaded rest, worker communication, clinical readiness and surveillance are distinct from the meteorological forecast and require responsible employers and public agencies. C. Forecast-triggered early action should precede severe impacts, but water-intensive cooling, indiscriminate groundwater extraction or unequal greening can increase future vulnerability and become maladaptation. D. Drought, water scarcity, heat, wildfire risk, crop stress, power demand and public-health pressure can interact, while humid heat or hot nights may intensify harm without changing the formal hazard label.

Answer: A. Explanation: Cool roofs, shade, ventilation, trees and water bodies can reduce selected exposures, but benefits depend on design, maintenance, local climate and access; a cooling intervention can transfer costs or water demand. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q62. Which option preserves the risk or institutional boundary of Urban heat measures?

A. A drought declaration, forecast, Heat Action Plan, dashboard, water scheme or contingency plan proves a classification or input; timely reach, equitable access, livelihood continuity and reduced illness require separate evidence. B. Cool roofs, shade, ventilation, trees and water bodies can reduce selected exposures, but benefits depend on design, maintenance, local climate and access; a cooling intervention can transfer costs or water demand. C. Forecast-triggered early action should precede severe impacts, but water-intensive cooling, indiscriminate groundwater extraction or unequal greening can increase future vulnerability and become maladaptation. D. Drought, water scarcity, heat, wildfire risk, crop stress, power demand and public-health pressure can interact, while humid heat or hot nights may intensify harm without changing the formal hazard label.

Answer: B. Explanation: Cool roofs, shade, ventilation, trees and water bodies can reduce selected exposures, but benefits depend on design, maintenance, local climate and access; a cooling intervention can transfer costs or water demand. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q63. Which statement uses Urban heat measures without changing its hazard, mandate or status?

A. Forecast-triggered early action should precede severe impacts, but water-intensive cooling, indiscriminate groundwater extraction or unequal greening can increase future vulnerability and become maladaptation. B. Aridity is a long-term climatic condition of low moisture availability, whereas drought is a temporary or episodic deficit relative to the expected local water regime. C. Cool roofs, shade, ventilation, trees and water bodies can reduce selected exposures, but benefits depend on design, maintenance, local climate and access; a cooling intervention can transfer costs or water demand. D. A drought declaration, forecast, Heat Action Plan, dashboard, water scheme or contingency plan proves a classification or input; timely reach, equitable access, livelihood continuity and reduced illness require separate evidence.

Answer: C. Explanation: Cool roofs, shade, ventilation, trees and water bodies can reduce selected exposures, but benefits depend on design, maintenance, local climate and access; a cooling intervention can transfer costs or water demand. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q64. Which option avoids the standard UPSC close-option trap about Urban heat measures?

A. Meteorological drought concerns prolonged inadequate or poorly distributed precipitation relative to the relevant climatology; it is a hazard indicator, not a complete account of impacts. B. A drought declaration, forecast, Heat Action Plan, dashboard, water scheme or contingency plan proves a classification or input; timely reach, equitable access, livelihood continuity and reduced illness require separate evidence. C. Aridity is a long-term climatic condition of low moisture availability, whereas drought is a temporary or episodic deficit relative to the expected local water regime. D. Cool roofs, shade, ventilation, trees and water bodies can reduce selected exposures, but benefits depend on design, maintenance, local climate and access; a cooling intervention can transfer costs or water demand.

Answer: D. Explanation: Cool roofs, shade, ventilation, trees and water bodies can reduce selected exposures, but benefits depend on design, maintenance, local climate and access; a cooling intervention can transfer costs or water demand. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q65. Which statement correctly identifies Labour and health measures?

A. Risk-sensitive work-rest scheduling, drinking water, shaded rest, worker communication, clinical readiness and surveillance are distinct from the meteorological forecast and require responsible employers and public agencies. B. Forecast-triggered early action should precede severe impacts, but water-intensive cooling, indiscriminate groundwater extraction or unequal greening can increase future vulnerability and become maladaptation. C. A drought declaration, forecast, Heat Action Plan, dashboard, water scheme or contingency plan proves a classification or input; timely reach, equitable access, livelihood continuity and reduced illness require separate evidence. D. Drought, water scarcity, heat, wildfire risk, crop stress, power demand and public-health pressure can interact, while humid heat or hot nights may intensify harm without changing the formal hazard label.

Answer: A. Explanation: Risk-sensitive work-rest scheduling, drinking water, shaded rest, worker communication, clinical readiness and surveillance are distinct from the meteorological forecast and require responsible employers and public agencies. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q66. Which option preserves the risk or institutional boundary of Labour and health measures?

A. A drought declaration, forecast, Heat Action Plan, dashboard, water scheme or contingency plan proves a classification or input; timely reach, equitable access, livelihood continuity and reduced illness require separate evidence. B. Risk-sensitive work-rest scheduling, drinking water, shaded rest, worker communication, clinical readiness and surveillance are distinct from the meteorological forecast and require responsible employers and public agencies. C. Aridity is a long-term climatic condition of low moisture availability, whereas drought is a temporary or episodic deficit relative to the expected local water regime. D. Forecast-triggered early action should precede severe impacts, but water-intensive cooling, indiscriminate groundwater extraction or unequal greening can increase future vulnerability and become maladaptation.

Answer: B. Explanation: Risk-sensitive work-rest scheduling, drinking water, shaded rest, worker communication, clinical readiness and surveillance are distinct from the meteorological forecast and require responsible employers and public agencies. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q67. Which statement uses Labour and health measures without changing its hazard, mandate or status?

A. Meteorological drought concerns prolonged inadequate or poorly distributed precipitation relative to the relevant climatology; it is a hazard indicator, not a complete account of impacts. B. A drought declaration, forecast, Heat Action Plan, dashboard, water scheme or contingency plan proves a classification or input; timely reach, equitable access, livelihood continuity and reduced illness require separate evidence. C. Risk-sensitive work-rest scheduling, drinking water, shaded rest, worker communication, clinical readiness and surveillance are distinct from the meteorological forecast and require responsible employers and public agencies. D. Aridity is a long-term climatic condition of low moisture availability, whereas drought is a temporary or episodic deficit relative to the expected local water regime.

Answer: C. Explanation: Risk-sensitive work-rest scheduling, drinking water, shaded rest, worker communication, clinical readiness and surveillance are distinct from the meteorological forecast and require responsible employers and public agencies. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q68. Which option avoids the standard UPSC close-option trap about Labour and health measures?

A. Agricultural drought arises when soil moisture and crop-water availability become inadequate for crops, so rainfall information must be read with sowing, vegetation and soil-moisture evidence. B. Aridity is a long-term climatic condition of low moisture availability, whereas drought is a temporary or episodic deficit relative to the expected local water regime. C. Meteorological drought concerns prolonged inadequate or poorly distributed precipitation relative to the relevant climatology; it is a hazard indicator, not a complete account of impacts. D. Risk-sensitive work-rest scheduling, drinking water, shaded rest, worker communication, clinical readiness and surveillance are distinct from the meteorological forecast and require responsible employers and public agencies.

Answer: D. Explanation: Risk-sensitive work-rest scheduling, drinking water, shaded rest, worker communication, clinical readiness and surveillance are distinct from the meteorological forecast and require responsible employers and public agencies. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q69. Which statement correctly identifies Compound risk?

A. Drought, water scarcity, heat, wildfire risk, crop stress, power demand and public-health pressure can interact, while humid heat or hot nights may intensify harm without changing the formal hazard label. B. A drought declaration, forecast, Heat Action Plan, dashboard, water scheme or contingency plan proves a classification or input; timely reach, equitable access, livelihood continuity and reduced illness require separate evidence. C. Aridity is a long-term climatic condition of low moisture availability, whereas drought is a temporary or episodic deficit relative to the expected local water regime. D. Forecast-triggered early action should precede severe impacts, but water-intensive cooling, indiscriminate groundwater extraction or unequal greening can increase future vulnerability and become maladaptation.

Answer: A. Explanation: Drought, water scarcity, heat, wildfire risk, crop stress, power demand and public-health pressure can interact, while humid heat or hot nights may intensify harm without changing the formal hazard label. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q70. Which option preserves the risk or institutional boundary of Compound risk?

A. Aridity is a long-term climatic condition of low moisture availability, whereas drought is a temporary or episodic deficit relative to the expected local water regime. B. Drought, water scarcity, heat, wildfire risk, crop stress, power demand and public-health pressure can interact, while humid heat or hot nights may intensify harm without changing the formal hazard label. C. A drought declaration, forecast, Heat Action Plan, dashboard, water scheme or contingency plan proves a classification or input; timely reach, equitable access, livelihood continuity and reduced illness require separate evidence. D. Meteorological drought concerns prolonged inadequate or poorly distributed precipitation relative to the relevant climatology; it is a hazard indicator, not a complete account of impacts.

Answer: B. Explanation: Drought, water scarcity, heat, wildfire risk, crop stress, power demand and public-health pressure can interact, while humid heat or hot nights may intensify harm without changing the formal hazard label. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q71. Which statement uses Compound risk without changing its hazard, mandate or status?

A. Aridity is a long-term climatic condition of low moisture availability, whereas drought is a temporary or episodic deficit relative to the expected local water regime. B. Agricultural drought arises when soil moisture and crop-water availability become inadequate for crops, so rainfall information must be read with sowing, vegetation and soil-moisture evidence. C. Drought, water scarcity, heat, wildfire risk, crop stress, power demand and public-health pressure can interact, while humid heat or hot nights may intensify harm without changing the formal hazard label. D. Meteorological drought concerns prolonged inadequate or poorly distributed precipitation relative to the relevant climatology; it is a hazard indicator, not a complete account of impacts.

Answer: C. Explanation: Drought, water scarcity, heat, wildfire risk, crop stress, power demand and public-health pressure can interact, while humid heat or hot nights may intensify harm without changing the

formal hazard label. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q72. Which option avoids the standard UPSC close-option trap about Compound risk?

A. Agricultural drought arises when soil moisture and crop-water availability become inadequate for crops, so rainfall information must be read with sowing, vegetation and soil-moisture evidence. B. Hydrological drought concerns below-normal availability in rivers, reservoirs, lakes or aquifers and may lag behind the rainfall deficit that initiated it. C. Meteorological drought concerns prolonged inadequate or poorly distributed precipitation relative to the relevant climatology; it is a hazard indicator, not a complete account of impacts. D. Drought, water scarcity, heat, wildfire risk, crop stress, power demand and public-health pressure can interact, while humid heat or hot nights may intensify harm without changing the formal hazard label.

Answer: D. Explanation: Drought, water scarcity, heat, wildfire risk, crop stress, power demand and public-health pressure can interact, while humid heat or hot nights may intensify harm without changing the formal hazard label. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q73. Which statement correctly identifies Early action and maladaptation?

A. Forecast-triggered early action should precede severe impacts, but water-intensive cooling, indiscriminate groundwater extraction or unequal greening can increase future vulnerability and become maladaptation. B. Aridity is a long-term climatic condition of low moisture availability, whereas drought is a temporary or episodic deficit relative to the expected local water regime. C. A drought declaration, forecast, Heat Action Plan, dashboard, water scheme or contingency plan proves a classification or input; timely reach, equitable access, livelihood continuity and reduced illness require separate evidence. D. Meteorological drought concerns prolonged inadequate or poorly distributed precipitation relative to the relevant climatology; it is a hazard indicator, not a complete account of impacts.

Answer: A. Explanation: Forecast-triggered early action should precede severe impacts, but water-intensive cooling, indiscriminate groundwater extraction or unequal greening can increase future vulnerability and become maladaptation. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q74. Which option preserves the risk or institutional boundary of Early action and maladaptation?

A. Meteorological drought concerns prolonged inadequate or poorly distributed precipitation relative to the relevant climatology; it is a hazard indicator, not a complete account of impacts. B. Forecast-triggered early action should precede severe impacts, but water-intensive cooling, indiscriminate groundwater extraction or unequal greening can increase future vulnerability and become maladaptation. C. Aridity is a long-term climatic condition of low moisture availability, whereas drought is a temporary or episodic deficit relative to the expected local water regime. D. Agricultural drought arises when soil moisture and crop-water availability become inadequate for crops, so rainfall information must be read with sowing, vegetation and soil-moisture evidence.

Answer: B. Explanation: Forecast-triggered early action should precede severe impacts, but water-intensive cooling, indiscriminate groundwater extraction or unequal greening can increase future vulnerability and become maladaptation. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q75. Which statement uses Early action and maladaptation without changing its hazard, mandate or status?

A. Meteorological drought concerns prolonged inadequate or poorly distributed precipitation relative to the relevant climatology; it is a hazard indicator, not a complete account of impacts. B. Agricultural drought arises when soil moisture and crop-water availability become inadequate for crops, so rainfall information must be read with sowing, vegetation and soil-moisture evidence. C. Forecast-triggered early action should precede severe impacts, but water-intensive cooling, indiscriminate groundwater extraction or unequal greening can increase future vulnerability and become maladaptation. D. Hydrological drought concerns below-normal availability in rivers, reservoirs, lakes or aquifers and may lag behind the rainfall deficit that initiated it.

Answer: C. Explanation: Forecast-triggered early action should precede severe impacts, but water-intensive cooling, indiscriminate groundwater extraction or unequal greening can increase future vulnerability and become maladaptation. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q76. Which option avoids the standard UPSC close-option trap about Early action and maladaptation?

A. Hydrological drought concerns below-normal availability in rivers, reservoirs, lakes or aquifers and may lag behind the rainfall deficit that initiated it. B. Socio-economic drought occurs when water shortage disrupts the supply of goods, services, livelihoods or essential uses; it expresses the interaction of physical scarcity with demand and unequal access. C. Agricultural drought arises when soil moisture and crop-water availability become inadequate for crops, so rainfall information must be read with sowing, vegetation and soil-moisture evidence. D. Forecast-triggered early action should precede severe impacts, but water-intensive cooling, indiscriminate groundwater extraction or unequal greening can increase future vulnerability and become maladaptation.

Answer: D. Explanation: Forecast-triggered early action should precede severe impacts, but water-intensive cooling, indiscriminate groundwater extraction or unequal greening can increase future vulnerability and become maladaptation. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q77. Which statement correctly identifies Monitoring-to-outcome firewall?

A. A drought declaration, forecast, Heat Action Plan, dashboard, water scheme or contingency plan proves a classification or input; timely reach, equitable access, livelihood continuity and reduced illness require separate evidence. B. Aridity is a long-term climatic condition of low moisture availability, whereas drought is a temporary or episodic deficit relative to the expected local water regime. C. Agricultural drought arises when soil moisture and crop-water availability become inadequate for crops, so rainfall information must be read with sowing, vegetation and soil-moisture evidence. D. Meteorological drought concerns prolonged inadequate or poorly distributed precipitation relative to the relevant climatology; it is a hazard indicator, not a complete account of impacts.

Answer: A. Explanation: A drought declaration, forecast, Heat Action Plan, dashboard, water scheme or contingency plan proves a classification or input; timely reach, equitable access, livelihood continuity and reduced illness require separate evidence. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q78. Which option preserves the risk or institutional boundary of Monitoring-to-outcome firewall?

A. Meteorological drought concerns prolonged inadequate or poorly distributed precipitation relative to the relevant climatology; it is a hazard indicator, not a complete account of impacts. B. A drought declaration, forecast, Heat Action Plan, dashboard, water scheme or contingency plan proves a classification or input; timely reach, equitable access, livelihood continuity and reduced illness require separate evidence. C. Agricultural drought arises when soil moisture and crop-water availability become inadequate for crops, so rainfall information must be read with sowing, vegetation and soil-moisture evidence. D. Hydrological drought concerns below-normal availability in rivers, reservoirs, lakes or aquifers and may lag behind the rainfall deficit that initiated it.

Answer: B. Explanation: A drought declaration, forecast, Heat Action Plan, dashboard, water scheme or contingency plan proves a classification or input; timely reach, equitable access, livelihood continuity and reduced illness require separate evidence. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q79. Which statement uses Monitoring-to-outcome firewall without changing its hazard, mandate or status?

A. Agricultural drought arises when soil moisture and crop-water availability become inadequate for crops, so rainfall information must be read with sowing, vegetation and soil-moisture evidence. B. Hydrological drought concerns below-normal availability in rivers, reservoirs, lakes or aquifers and may lag behind the rainfall deficit that initiated it. C. A drought declaration, forecast, Heat Action Plan, dashboard, water scheme or contingency plan proves a classification or input; timely reach, equitable access, livelihood continuity and reduced illness require separate evidence. D. Socio-economic drought occurs when water shortage disrupts the supply of goods, services, livelihoods or essential uses; it expresses the interaction of physical scarcity with demand and unequal access.

Answer: C. Explanation: A drought declaration, forecast, Heat Action Plan, dashboard, water scheme or contingency plan proves a classification or input; timely reach, equitable access, livelihood continuity and reduced illness require separate evidence. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q80. Which option avoids the standard UPSC close-option trap about Monitoring-to-outcome firewall?

A. Socio-economic drought occurs when water shortage disrupts the supply of goods, services, livelihoods or essential uses; it expresses the interaction of physical scarcity with demand and unequal access. B. Hydrological drought concerns below-normal availability in rivers, reservoirs, lakes or aquifers and may lag behind the rainfall deficit that initiated it. C. Ecological drought describes water shortage severe enough to impair ecosystem productivity or function; it must not be silently substituted for socio-economic drought. D. A drought declaration, forecast, Heat Action Plan, dashboard, water scheme or contingency plan proves a classification or input; timely reach, equitable access, livelihood continuity and reduced illness require separate evidence.

Answer: D. Explanation: A drought declaration, forecast, Heat Action Plan, dashboard, water scheme or contingency plan proves a classification or input; timely reach, equitable access, livelihood continuity and reduced illness require separate evidence. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

PYQS AND ANSWER PRACTICE

AUDITED TOPIC-SPECIFIC PYQ OWNERSHIP

The 2025 wet-bulb card is the only direct topic-specific route. The 2024 resilience and 2020 proactive-governance cards are explicitly support/adjacent applications and do not manufacture drought- or heat-specific PYQs.

PYQ DEMAND CARD 1 - 2025 Prelims GS-I

Demand: Assess the implications of wet-bulb temperature crossing the stated benchmark.

Status: Verified direct objective route; the official Set-A key exists locally, but this card records no answer option and preserves the wet-bulb-versus-IMD-threshold distinction.

Model solution: Heat-wave hazard: A heat wave is a meteorological hazard identified through IMD's station and departure or actual-temperature criteria; it is not synonymous with every hot day or every episode of heat illness. **Heat stress:** Heat stress is the physiological burden produced by temperature, humidity, radiation, wind, exertion, clothing, health and access to rest, water or cooling; exposure and vulnerability therefore mediate the hazard. **Wet-bulb boundary:** Wet-bulb temperature combines heat and humidity by representing evaporative-cooling potential; it is not an IMD heat-wave declaration threshold, and dangerous stress can occur below a theoretical upper-limit benchmark. **Exposure inequality:** Outdoor and informal workers, older persons, children, people with illness, poorly housed households and those lacking dependable water or cooling may face different risk under the same forecast temperature. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Demand decoding: Treat ‘PYQ DEMAND CARD 1 - 2025 Prelims GS-I' as a hazard/risk term, legal-status, institutional-mandate, cycle-stage, warning-stage, date and evidence problem.

Detailed examiner-grade model answer:

Introduction and thesis: Heat-wave hazard: A heat wave is a meteorological hazard identified through IMD's station and departure or actual-temperature criteria; it is not synonymous with every hot day or every episode of heat illness. **Heat stress:** Heat stress is the physiological burden produced by temperature, humidity, radiation, wind, exertion, clothing, health and access to rest, water or cooling; exposure and vulnerability therefore mediate the hazard. **Wet-bulb boundary:** Wet-bulb temperature combines heat and humidity by representing evaporative-cooling potential; it is not an IMD heat-wave declaration threshold, and dangerous stress can occur below a theoretical upper-limit benchmark. **Exposure inequality:** Outdoor and informal workers, older persons, children, people with illness, poorly housed households and those lacking dependable water or cooling may face different risk under the same forecast temperature. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Analytical body:

1. **Claim:** Demand: Assess the implications of wet-bulb temperature crossing the stated benchmark. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.
2. **Claim:** Status: Verified direct objective route; the official Set-A key exists locally, but this card records no answer option and preserves the wet-bulb-versus-IMD-threshold distinction. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.

Counter-position / limit: An Act, plan, guideline, scheme, forecast, warning, drill, team, sanctioned capacity, allocation or dispatch cannot alone establish implementation, evacuation, expenditure, reduced loss, equitable recovery or resilience; test mandate, process, status and evidence.

Qualified conclusion: Heat-wave hazard: A heat wave is a meteorological hazard identified through IMD's station and departure or actual-temperature criteria; it is not synonymous with every hot day or every episode of heat illness. **Heat stress:** Heat stress is the physiological burden produced by temperature, humidity, radiation, wind, exertion, clothing, health and access to rest, water or cooling; exposure and vulnerability therefore mediate the hazard. **Wet-bulb boundary:** Wet-bulb temperature combines heat and humidity by representing evaporative-cooling potential; it is not an IMD heat-wave declaration threshold, and dangerous stress can occur below a theoretical upper-limit benchmark. **Exposure inequality:** Outdoor and informal workers, older persons, children, people with illness, poorly housed households and those lacking dependable water or cooling may face different risk under the same forecast temperature. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Executable exam-length answer / compression plan: Define the hazard and exposed system; separate vulnerability, capacity and impact; fix Act/rule/plan/guideline or institutional mandate; test warning, response, finance and outcome status; reject the nearest date, unit or jurisdiction trap.

Why this earns marks: It prevents hazard from impact, forecast from warning, NDMA from NDRF, capacity from deployment, relief norm from entitlement, and allocation from expenditure being conflated.

How to improve this answer: For ‘PYQ DEMAND CARD 1 - 2025 Prelims GS-I', explain why the nearest distractor fails on risk component, mandate, legal character, cycle stage, warning stage, unit, jurisdiction, status or date.

PYQ DEMAND CARD 2 - 2024 GS-III

Demand: Describe disaster resilience, its determination and the Sendai Framework elements.

Status: Verified support route, not a drought-specific PYQ; use slow-onset monitoring, early action and outcome evidence only as a bounded illustration.

Model solution: Slow-onset cascade: Drought can unfold from rainfall deficit through soil-moisture stress, storage decline, ecosystem damage, livelihood loss and health effects over different time scales, so one indicator cannot represent every stage. **Heat Action Plan:** A Heat Action Plan links seasonal preparedness, warnings, health-system readiness, public communication, water and cooling access, work or school timing and longer-term urban measures through locally assigned responsibilities. **Compound risk:** Drought, water scarcity, heat, wildfire risk, crop stress, power demand and public-health pressure can interact, while humid heat or hot nights may intensify harm without changing the formal hazard label. **Early action and maladaptation:** Forecast-triggered early action should precede severe impacts, but water-intensive cooling, indiscriminate groundwater extraction or unequal greening can increase future vulnerability and become maladaptation. **Monitoring-to-outcome firewall:** A drought declaration, forecast, Heat Action Plan, dashboard, water scheme or contingency plan proves a classification or input; timely reach, equitable access, livelihood continuity and reduced illness require separate evidence. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Demand decoding: The directive **answer** requires a direct position on 'PYQ DEMAND CARD 2 - 2024 GS-III', each clause, risk mechanism, named Indian law/institution/event, dated status, implementation and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Slow-onset cascade: Drought can unfold from rainfall deficit through soil-moisture stress, storage decline, ecosystem damage, livelihood loss and health effects over different time scales, so one indicator cannot represent every stage. **Heat Action Plan:** A Heat Action Plan links seasonal preparedness, warnings, health-system readiness, public communication, water and cooling access, work or school timing and longer-term urban measures through locally assigned responsibilities. **Compound risk:** Drought, water scarcity, heat, wildfire risk, crop stress, power demand and public-health pressure can interact, while humid heat or hot nights may intensify harm without changing the formal hazard label. **Early action and maladaptation:** Forecast-triggered early action should precede severe impacts, but water-intensive cooling, indiscriminate groundwater extraction or unequal greening can increase future vulnerability and become maladaptation. **Monitoring-to-outcome firewall:** A drought declaration, forecast, Heat Action Plan, dashboard, water scheme or contingency plan proves a classification or input; timely reach, equitable access, livelihood continuity and reduced illness require separate evidence. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Analytical body:

1. **Claim:** Demand: Describe disaster resilience, its determination and the Sendai Framework elements. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.
2. **Claim:** Status: Verified support route, not a drought-specific PYQ; use slow-onset monitoring, early action and outcome evidence only as a bounded illustration. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.

Counter-position / limit: An Act, plan, guideline, scheme, forecast, warning, drill, team, sanctioned capacity, allocation or dispatch cannot alone establish implementation, evacuation, expenditure, reduced loss, equitable recovery or resilience; test mandate, process, status and evidence.

Qualified conclusion: Slow-onset cascade: Drought can unfold from rainfall deficit through soil-moisture stress, storage decline, ecosystem damage, livelihood loss and health effects over different time scales, so one indicator cannot represent every stage. **Heat Action Plan:** A Heat Action Plan links seasonal preparedness, warnings, health-system readiness, public communication, water and cooling access, work or school timing and longer-term urban measures through locally assigned responsibilities. **Compound risk:** Drought, water scarcity, heat, wildfire risk, crop stress, power demand and public-health pressure can interact, while humid heat or hot nights may intensify harm without changing the formal hazard label. **Early action and maladaptation:** Forecast-triggered early action should precede severe impacts, but water-intensive cooling, indiscriminate groundwater extraction or unequal greening can increase future vulnerability and become maladaptation. **Monitoring-to-outcome firewall:** A drought declaration, forecast, Heat Action Plan, dashboard, water scheme or contingency plan proves a classification or input; timely reach, equitable access, livelihood continuity and reduced illness require separate evidence. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Executable exam-length answer / compression plan: For 15 marks, spend one-sixth of the time decoding the directive and drawing hazard -> exposure/vulnerability/capacity -> risk/impact -> prevention/mitigation/preparedness -> response/recovery/BBB; write four to seven claim -> named evidence/example -> analysis -> qualification points; reserve the final minute for source, date, unit, status, mandate and event/inference checks.

Why this earns marks: The answer obeys the directive, uses India-centric evidence, and preserves risk terms, legal character, institutional mandate, warning/cycle stage, finance status, implementation and resilience distinctions.

How to improve this answer: For 'PYQ DEMAND CARD 2 - 2024 GS-III', replace the weakest generic point with one exact Indian mechanism, institution/guideline/event, dated source and unit, implementation bottleneck, local-capacity or inclusion safeguard and answer-specific qualification.

PYQ DEMAND CARD 3 - 2020 GS-III

Demand: Discuss the shift from reactive to proactive disaster management in India.

Status: Verified adjacent governance route; drought and heat supply an anticipatory-action example, not a claim that the printed question named either hazard.

Model solution: Slow-onset cascade: Drought can unfold from rainfall deficit through soil-moisture stress, storage decline, ecosystem damage, livelihood loss and health effects over different time scales, so one indicator cannot represent every stage. **Multi-indicator declaration:** India's Manual for Drought Management uses rainfall, vegetative indices, crop-sowing progression, soil moisture and hydrological indices in a graded assessment rather than one universal crop-loss percentage. **Heat Action Plan:** A Heat Action Plan links seasonal preparedness, warnings, health-system readiness, public communication, water and cooling access, work or school timing and longer-term urban measures through locally assigned responsibilities. **Early action and maladaptation:** Forecast-triggered early action should precede severe impacts, but water-intensive cooling, indiscriminate groundwater extraction or unequal greening can increase future vulnerability and become maladaptation. **Monitoring-to-outcome firewall:** A drought declaration, forecast, Heat Action Plan, dashboard, water scheme or contingency plan proves a classification or input; timely reach, equitable access, livelihood continuity and reduced illness require separate evidence. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Demand decoding: The directive **answer** requires a direct position on 'PYQ DEMAND CARD 3 - 2020 GS-III', each clause, risk mechanism, named Indian law/institution/event, dated status, implementation and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Slow-onset cascade: Drought can unfold from rainfall deficit through soil-moisture stress, storage decline, ecosystem damage, livelihood loss and health effects over different time scales, so one indicator cannot represent every stage. **Multi-indicator declaration:** India's Manual for Drought Management uses rainfall, vegetative indices, crop-sowing progression, soil moisture and hydrological indices in a graded assessment rather than one universal crop-loss percentage. **Heat Action Plan:** A Heat Action Plan links seasonal preparedness, warnings, health-system readiness, public communication, water and cooling access, work or school timing and longer-term urban measures through locally assigned responsibilities. **Early action and maladaptation:** Forecast-triggered early action should precede severe impacts, but water-intensive cooling, indiscriminate groundwater extraction or unequal greening can increase future vulnerability and become maladaptation. **Monitoring-to-outcome firewall:** A drought declaration, forecast, Heat Action Plan, dashboard, water scheme or contingency plan proves a classification or input; timely reach, equitable access, livelihood continuity and reduced illness require separate evidence. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Analytical body:

1. **Claim:** Demand: Discuss the shift from reactive to proactive disaster management in India. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.
2. **Claim:** Status: Verified adjacent governance route; drought and heat supply an anticipatory-action example, not a claim that the printed question named either hazard. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.

Counter-position / limit: An Act, plan, guideline, scheme, forecast, warning, drill, team, sanctioned capacity, allocation or dispatch cannot alone establish implementation, evacuation, expenditure, reduced loss, equitable recovery or resilience; test mandate, process, status and evidence.

Qualified conclusion: Slow-onset cascade: Drought can unfold from rainfall deficit through soil-moisture stress, storage decline, ecosystem damage, livelihood loss and health effects over different time scales, so one indicator cannot represent every stage. **Multi-indicator declaration:** India's Manual for Drought Management uses rainfall, vegetative indices, crop-sowing progression, soil moisture and hydrological indices in a graded assessment rather than one universal crop-loss percentage. **Heat Action Plan:** A Heat Action Plan links seasonal preparedness, warnings, health-system readiness, public communication, water and cooling access, work or school timing and longer-term urban measures through locally assigned responsibilities. **Early action and maladaptation:** Forecast-triggered early action should precede severe impacts, but water-intensive cooling, indiscriminate groundwater extraction or unequal greening can increase future vulnerability and become maladaptation. **Monitoring-to-outcome firewall:** A drought declaration, forecast, Heat Action Plan, dashboard, water scheme or contingency plan proves a classification or input; timely reach, equitable access, livelihood continuity and reduced illness require separate evidence. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Executable exam-length answer / compression plan: For 15 marks, spend one-sixth of the time decoding the directive and drawing hazard -> exposure/vulnerability/capacity -> risk/impact -> prevention/mitigation/preparedness -> response/recovery/BBB; write four to seven claim -> named evidence/example -> analysis -> qualification points; reserve the final minute for source, date, unit, status, mandate and event/inference checks.

Why this earns marks: The answer obeys the directive, uses India-centric evidence, and preserves risk terms, legal character, institutional mandate, warning/cycle stage, finance status, implementation and resilience distinctions.

How to improve this answer: For ‘PYQ DEMAND CARD 3 - 2020 GS-III', replace the weakest generic point with one exact Indian mechanism, institution/guideline/event, dated source and unit, implementation bottleneck, local-capacity or inclusion safeguard and answer-specific qualification.

ORIGINAL MAINS 1 - 10 MARKS

Question: Distinguish aridity, water scarcity and the principal drought types. Answer in about 150 words.

Model thesis: Claim: Aridity versus drought. **Named evidence/example:** Aridity is a long-term climatic condition of low moisture availability, whereas drought is a temporary or episodic deficit relative to the expected local water regime. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Meteorological drought. **Named evidence/example:** Meteorological drought concerns prolonged inadequate or poorly distributed precipitation relative to the relevant climatology; it is a hazard indicator, not a complete account of impacts. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Agricultural drought. **Named evidence/example:** Agricultural drought arises when soil moisture and crop-water availability become inadequate for crops, so rainfall information must be read with sowing, vegetation and soil-moisture evidence. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Hydrological drought. **Named evidence/example:** Hydrological drought concerns below-normal availability in rivers, reservoirs, lakes or aquifers and may lag behind the rainfall deficit that initiated it. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Socio-economic drought. **Named evidence/example:** Socio-economic drought occurs when water shortage disrupts the supply of goods, services, livelihoods or essential uses; it expresses the interaction of physical scarcity with demand and unequal access. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Ecological drought. **Named evidence/example:** Ecological drought describes water shortage severe enough to impair ecosystem productivity or function; it must not be silently substituted for socio-economic drought. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

Claim -> named evidence -> analysis -> qualification:

- Aridity is a long-term climatic condition of low moisture availability, whereas drought is a temporary or episodic deficit relative to the expected local water regime.
- Meteorological drought concerns prolonged inadequate or poorly distributed precipitation relative to the relevant climatology; it is a hazard indicator, not a complete account of impacts.
- Agricultural drought arises when soil moisture and crop-water availability become inadequate for crops, so rainfall information must be read with sowing, vegetation and soil-moisture evidence.
- Hydrological drought concerns below-normal availability in rivers, reservoirs, lakes or aquifers and may lag behind the rainfall deficit that initiated it.
- Socio-economic drought occurs when water shortage disrupts the supply of goods, services, livelihoods or essential uses; it expresses the interaction of physical scarcity with demand and unequal access.
- Ecological drought describes water shortage severe enough to impair ecosystem productivity or function; it must not be silently substituted for socio-economic drought.

Qualified conclusion: Claim: Aridity versus drought. **Named evidence/example:** Aridity is a long-term climatic condition of low moisture availability, whereas drought is a temporary or episodic deficit relative to the expected local water regime. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Meteorological drought. **Named evidence/example:** Meteorological drought concerns prolonged inadequate or poorly distributed precipitation relative to the relevant climatology; it is a hazard indicator, not a complete account of impacts. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Agricultural drought. **Named evidence/example:** Agricultural drought arises when soil moisture and crop-water availability become inadequate for crops, so rainfall information must be read with sowing, vegetation and soil-moisture evidence. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Hydrological drought. **Named evidence/example:** Hydrological drought concerns below-normal availability in rivers, reservoirs, lakes or aquifers and may lag behind the rainfall deficit that initiated it. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Socio-economic drought. **Named evidence/example:** Socio-economic drought occurs when water shortage disrupts the supply of goods, services, livelihoods or essential uses; it expresses the interaction of physical scarcity with demand and unequal access. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Ecological drought. **Named evidence/example:** Ecological drought describes water shortage severe enough to impair ecosystem productivity or function; it must not be silently substituted for socio-economic drought. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

Demand decoding: The directive **answer** requires a direct position on ‘Distinguish aridity, water scarcity and the principal drought types. Answer in about 150...', each clause, risk mechanism, named Indian law/institution/event, dated status, implementation and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Aridity versus drought. **Named evidence/example:** Aridity is a long-term climatic condition of low moisture availability, whereas drought is a temporary or episodic deficit relative to the expected local water regime. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Meteorological drought. **Named evidence/example:** Meteorological drought concerns prolonged inadequate or poorly distributed precipitation relative to the relevant climatology; it is a hazard indicator, not a complete account of impacts. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Agricultural drought. **Named evidence/example:** Agricultural drought arises when soil moisture and crop-water availability become inadequate for crops, so rainfall information must be read with sowing, vegetation and soil-moisture evidence. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Hydrological drought. **Named evidence/example:** Hydrological drought concerns below-normal availability in rivers, reservoirs, lakes or aquifers and may lag behind the rainfall deficit that initiated it. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Socio-economic drought. **Named evidence/example:** Socio-economic drought occurs when water shortage disrupts the supply of goods, services, livelihoods or essential uses; it expresses the interaction of physical scarcity with demand and unequal access. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Ecological drought. **Named evidence/example:** Ecological drought describes water shortage severe enough to impair ecosystem productivity or function; it must not be silently substituted for socio-economic drought. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

Analytical body:

1. **Claim:** Aridity is a long-term climatic condition of low moisture availability, whereas drought is a temporary or episodic deficit relative to the expected local water regime. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.
2. **Claim:** Meteorological drought concerns prolonged inadequate or poorly distributed precipitation relative to the relevant climatology; it is a hazard indicator, not a complete account of impacts. **Named evidence/example:**

Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.

3. **Claim:** Agricultural drought arises when soil moisture and crop-water availability become inadequate for crops, so rainfall information must be read with sowing, vegetation and soil-moisture evidence. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.
4. **Claim:** Hydrological drought concerns below-normal availability in rivers, reservoirs, lakes or aquifers and may lag behind the rainfall deficit that initiated it. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.
5. **Claim:** Socio-economic drought occurs when water shortage disrupts the supply of goods, services, livelihoods or essential uses; it expresses the interaction of physical scarcity with demand and unequal access. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.
6. **Claim:** Ecological drought describes water shortage severe enough to impair ecosystem productivity or function; it must not be silently substituted for socio-economic drought. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.

Counter-position / limit: An Act, plan, guideline, scheme, forecast, warning, drill, team, sanctioned capacity, allocation or dispatch cannot alone establish implementation, evacuation, expenditure, reduced loss, equitable recovery or resilience; test mandate, process, status and evidence.

Qualified conclusion: Claim: Aridity versus drought. **Named evidence/example:** Aridity is a long-term climatic condition of low moisture availability, whereas drought is a temporary or episodic deficit relative to the expected local water regime. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Meteorological drought. **Named evidence/example:** Meteorological drought concerns prolonged inadequate or poorly distributed precipitation relative to the relevant climatology; it is a hazard indicator, not a complete account of impacts. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Agricultural drought. **Named evidence/example:** Agricultural drought arises when soil moisture and crop-water availability become inadequate for crops, so rainfall information must be read with sowing, vegetation and soil-moisture evidence. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Hydrological drought. **Named evidence/example:** Hydrological drought concerns below-normal availability in rivers, reservoirs, lakes or aquifers and may lag behind the rainfall deficit that initiated it. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Socio-economic drought. **Named evidence/example:** Socio-economic drought occurs when water shortage disrupts the supply of goods, services, livelihoods or essential uses; it expresses the interaction of physical scarcity with demand and unequal access. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Ecological drought. **Named evidence/example:** Ecological drought describes water shortage severe enough to impair ecosystem productivity or function; it must not be silently substituted for socio-economic drought. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

Executable exam-length answer / compression plan: For 10 marks, spend one-sixth of the time decoding the directive and drawing hazard -> exposure/vulnerability/capacity -> risk/impact -> prevention/mitigation/preparedness -> response/recovery/BBB; write four to seven claim -> named evidence/example -> analysis -> qualification points; reserve the final minute for source, date, unit, status, mandate and event/inference checks.

Why this earns marks: The answer obeys the directive, uses India-centric evidence, and preserves risk terms, legal character, institutional mandate, warning/cycle stage, finance status, implementation and resilience distinctions.

How to improve this answer: For ‘Distinguish aridity, water scarcity and the principal drought types. Answer in about 150...', replace the weakest generic point with one exact Indian mechanism, institution/guideline/event, dated source and unit, implementation bottleneck, local-capacity or inclusion safeguard and answer-specific qualification.

ORIGINAL MAINS 2 - 10 MARKS

Question: Explain why heat-wave hazard and heat stress are not interchangeable. Answer in about 150 words.

Model thesis: Claim: Heat-wave hazard. **Named evidence/example:** A heat wave is a meteorological hazard identified through IMD's station and departure or actual-temperature criteria; it is not synonymous with every hot day or every episode of heat illness. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Heat stress. **Named evidence/example:** Heat stress is the physiological burden produced by temperature, humidity, radiation, wind, exertion, clothing, health and access to rest, water or cooling; exposure and vulnerability therefore mediate the hazard. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Wet-bulb boundary. **Named evidence/example:** Wet-bulb temperature combines heat and humidity by representing evaporative-cooling potential; it is not an IMD heat-wave declaration threshold, and dangerous stress can occur below a theoretical upper-limit benchmark. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Exposure inequality. **Named evidence/example:** Outdoor and informal workers, older persons, children, people with illness, poorly housed households and those lacking dependable water or cooling may face different risk under the same forecast temperature. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

Claim -> named evidence -> analysis -> qualification:

- A heat wave is a meteorological hazard identified through IMD's station and departure or actual-temperature criteria; it is not synonymous with every hot day or every episode of heat illness.
- Heat stress is the physiological burden produced by temperature, humidity, radiation, wind, exertion, clothing, health and access to rest, water or cooling; exposure and vulnerability therefore mediate the hazard.
- Wet-bulb temperature combines heat and humidity by representing evaporative-cooling potential; it is not an IMD heat-wave declaration threshold, and dangerous stress can occur below a theoretical upper-limit benchmark.
- Outdoor and informal workers, older persons, children, people with illness, poorly housed households and those lacking dependable water or cooling may face different risk under the same forecast temperature.

Qualified conclusion: Claim: Heat-wave hazard. **Named evidence/example:** A heat wave is a meteorological hazard identified through IMD's station and departure or actual-temperature criteria; it is not synonymous with every hot day or every episode of heat illness. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Heat stress. **Named evidence/example:** Heat stress is the physiological burden produced by temperature, humidity, radiation, wind, exertion, clothing, health and access to rest, water or cooling; exposure and vulnerability therefore mediate the hazard. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Wet-bulb boundary. **Named evidence/example:** Wet-bulb temperature combines heat and humidity by representing evaporative-cooling potential; it is not an IMD heat-wave declaration threshold, and dangerous stress can occur below a theoretical upper-limit benchmark. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Exposure inequality. **Named evidence/example:** Outdoor and informal workers, older persons, children, people with illness, poorly housed households and those lacking dependable water or cooling may face different risk under the same forecast temperature. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

Demand decoding: The directive **explain** requires a direct position on ‘Explain why heat-wave hazard and heat stress are not interchangeable. Answer in about 150...', each clause, risk mechanism, named Indian law/institution/event, dated status, implementation and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Heat-wave hazard. **Named evidence/example:** A heat wave is a meteorological hazard identified through IMD's station and departure or actual-temperature criteria; it is not synonymous with every hot day or every episode of heat illness. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Heat stress. **Named evidence/example:** Heat stress is the physiological burden produced by temperature, humidity, radiation, wind, exertion, clothing, health and access to rest, water or cooling; exposure and vulnerability therefore mediate the hazard. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Wet-bulb boundary. **Named evidence/example:** Wet-bulb temperature combines heat and humidity by representing evaporative-cooling potential; it is not an IMD heat-wave declaration threshold, and dangerous stress can occur below a theoretical upper-limit benchmark. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Exposure inequality. **Named evidence/example:** Outdoor and informal workers, older persons, children, people with illness, poorly housed households and those lacking dependable water or cooling may face different risk under the same forecast temperature. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

Analytical body:

1. **Claim:** A heat wave is a meteorological hazard identified through IMD's station and departure or actual-temperature criteria; it is not synonymous with every hot day or every episode of heat illness. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.

2. **Claim:** Heat stress is the physiological burden produced by temperature, humidity, radiation, wind, exertion, clothing, health and access to rest, water or cooling; exposure and vulnerability therefore mediate the hazard. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.
3. **Claim:** Wet-bulb temperature combines heat and humidity by representing evaporative-cooling potential; it is not an IMD heat-wave declaration threshold, and dangerous stress can occur below a theoretical upper-limit benchmark. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.
4. **Claim:** Outdoor and informal workers, older persons, children, people with illness, poorly housed households and those lacking dependable water or cooling may face different risk under the same forecast temperature. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.

Counter-position / limit: An Act, plan, guideline, scheme, forecast, warning, drill, team, sanctioned capacity, allocation or dispatch cannot alone establish implementation, evacuation, expenditure, reduced loss, equitable recovery or resilience; test mandate, process, status and evidence.

Qualified conclusion: Claim: Heat-wave hazard. **Named evidence/example:** A heat wave is a meteorological hazard identified through IMD's station and departure or actual-temperature criteria; it is not synonymous with every hot day or every episode of heat illness. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Heat stress. **Named evidence/example:** Heat stress is the physiological burden produced by temperature, humidity, radiation, wind, exertion, clothing, health and access to rest, water or cooling; exposure and vulnerability therefore mediate the hazard. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Wet-bulb boundary. **Named evidence/example:** Wet-bulb temperature combines heat and humidity by representing evaporative-cooling potential; it is not an IMD heat-wave declaration threshold, and dangerous stress can occur below a theoretical upper-limit benchmark. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Exposure inequality. **Named evidence/example:** Outdoor and informal workers, older persons, children, people with illness, poorly housed households and those lacking dependable water or cooling may face different risk under the same forecast temperature. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

Executable exam-length answer / compression plan: For 10 marks, spend one-sixth of the time decoding the directive and drawing hazard -> exposure/vulnerability/capacity -> risk/impact -> prevention/mitigation/preparedness -> response/recovery/BBB; write four to seven claim -> named evidence/example -> analysis -> qualification points; reserve the final minute for source, date, unit, status, mandate and event/inference checks.

Why this earns marks: The answer obeys the directive, uses India-centric evidence, and preserves risk terms, legal character, institutional mandate, warning/cycle stage, finance status, implementation and resilience distinctions.

How to improve this answer: For ‘Explain why heat-wave hazard and heat stress are not interchangeable. Answer in about 150...', replace the weakest generic point with one exact Indian mechanism, institution/guideline/event, dated source and unit, implementation bottleneck, local-capacity or inclusion safeguard and answer-specific qualification.

ORIGINAL MAINS 3 - 15 MARKS

Question: Analyse drought as a slow-onset cascade requiring multi-indicator monitoring and early action. Answer in about 250 words.

Model thesis: Claim: Meteorological drought. **Named evidence/example:** Meteorological drought concerns prolonged inadequate or poorly distributed precipitation relative to the relevant climatology; it is a hazard indicator, not a complete account of impacts. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Agricultural drought. **Named evidence/example:** Agricultural drought arises when soil moisture and crop-water availability become inadequate for crops, so rainfall information must be read with sowing, vegetation and soil-moisture evidence. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Hydrological drought. **Named evidence/example:** Hydrological drought concerns below-normal availability in rivers, reservoirs, lakes or aquifers and may lag behind the rainfall deficit that initiated it. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Socio-economic drought. **Named evidence/example:** Socio-economic drought occurs when water shortage disrupts the supply of goods, services, livelihoods or essential uses; it expresses the interaction of physical scarcity with demand and unequal access. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Slow-onset cascade. **Named evidence/example:** Drought can unfold from rainfall deficit through soil-moisture stress, storage decline, ecosystem damage, livelihood loss and health effects over different time scales, so one indicator cannot represent every stage. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Multi-indicator declaration. **Named evidence/example:** India's Manual for Drought Management uses rainfall, vegetative indices, crop-sowing progression, soil moisture and hydrological indices in a graded assessment rather than one universal crop-loss percentage. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Early action and maladaptation. **Named evidence/example:** Forecast-triggered early action should precede severe impacts, but water-intensive cooling, indiscriminate groundwater extraction or unequal greening can increase future vulnerability and become maladaptation. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

Claim -> named evidence -> analysis -> qualification:

- Meteorological drought concerns prolonged inadequate or poorly distributed precipitation relative to the relevant climatology; it is a hazard indicator, not a complete account of impacts.
- Agricultural drought arises when soil moisture and crop-water availability become inadequate for crops, so rainfall information must be read with sowing, vegetation and soil-moisture evidence.
- Hydrological drought concerns below-normal availability in rivers, reservoirs, lakes or aquifers and may lag behind the rainfall deficit that initiated it.
- Socio-economic drought occurs when water shortage disrupts the supply of goods, services, livelihoods or essential uses; it expresses the interaction of physical scarcity with demand and unequal access.
- Drought can unfold from rainfall deficit through soil-moisture stress, storage decline, ecosystem damage, livelihood loss and health effects over different time scales, so one indicator cannot represent every stage.
- India's Manual for Drought Management uses rainfall, vegetative indices, crop-sowing progression, soil moisture and hydrological indices in a graded assessment rather than one universal crop-loss percentage.

- Forecast-triggered early action should precede severe impacts, but water-intensive cooling, indiscriminate groundwater extraction or unequal greening can increase future vulnerability and become maladaptation.

Qualified conclusion: Claim: Meteorological drought. **Named evidence/example:** Meteorological drought concerns prolonged inadequate or poorly distributed precipitation relative to the relevant climatology; it is a hazard indicator, not a complete account of impacts. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Agricultural drought. **Named evidence/example:** Agricultural drought arises when soil moisture and crop-water availability become inadequate for crops, so rainfall information must be read with sowing, vegetation and soil-moisture evidence. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Hydrological drought. **Named evidence/example:** Hydrological drought concerns below-normal availability in rivers, reservoirs, lakes or aquifers and may lag behind the rainfall deficit that initiated it. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Socio-economic drought. **Named evidence/example:** Socio-economic drought occurs when water shortage disrupts the supply of goods, services, livelihoods or essential uses; it expresses the interaction of physical scarcity with demand and unequal access. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Slow-onset cascade. **Named evidence/example:** Drought can unfold from rainfall deficit through soil-moisture stress, storage decline, ecosystem damage, livelihood loss and health effects over different time scales, so one indicator cannot represent every stage. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Multi-indicator declaration. **Named evidence/example:** India's Manual for Drought Management uses rainfall, vegetative indices, crop-sowing progression, soil moisture and hydrological indices in a graded assessment rather than one universal crop-loss percentage. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Early action and maladaptation. **Named evidence/example:** Forecast-triggered early action should precede severe impacts, but water-intensive cooling, indiscriminate groundwater extraction or unequal greening can increase future vulnerability and become maladaptation. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

Demand decoding: The directive **analyse** requires a direct position on ‘Analyse drought as a slow-onset cascade requiring multi-indicator monitoring and early...', each clause, risk mechanism, named Indian law/institution/event, dated status, implementation and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Meteorological drought. **Named evidence/example:** Meteorological drought concerns prolonged inadequate or poorly distributed precipitation relative to the relevant climatology; it is a hazard indicator, not a complete account of impacts. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Agricultural drought. **Named evidence/example:** Agricultural drought arises when soil moisture and crop-water availability become inadequate for crops, so rainfall information must be read with sowing, vegetation and soil-moisture evidence. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Hydrological drought. **Named evidence/example:** Hydrological drought concerns below-normal availability in rivers, reservoirs, lakes or aquifers and may lag behind the rainfall deficit that initiated it. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Socio-economic drought. **Named evidence/example:** Socio-economic drought occurs when water shortage disrupts the supply of goods, services, livelihoods or essential uses; it expresses the interaction of physical scarcity with demand and unequal access. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Slow-onset cascade. **Named evidence/example:** Drought can unfold from rainfall deficit through soil-moisture stress, storage decline, ecosystem damage, livelihood loss and health effects over different time scales, so one indicator cannot represent every stage. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Multi-indicator declaration. **Named evidence/example:** India's Manual for Drought Management uses rainfall, vegetative indices, crop-sowing progression, soil moisture and hydrological indices in a graded assessment rather than one universal crop-loss percentage. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Early action and maladaptation. **Named evidence/example:** Forecast-triggered early action should precede severe impacts, but water-intensive cooling, indiscriminate groundwater extraction or unequal greening can increase future vulnerability and become maladaptation. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

Analytical body:

1. **Claim:** Meteorological drought concerns prolonged inadequate or poorly distributed precipitation relative to the relevant climatology; it is a hazard indicator, not a complete account of impacts. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.
2. **Claim:** Agricultural drought arises when soil moisture and crop-water availability become inadequate for crops, so rainfall information must be read with sowing, vegetation and soil-moisture evidence. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.
3. **Claim:** Hydrological drought concerns below-normal availability in rivers, reservoirs, lakes or aquifers and may lag behind the rainfall deficit that initiated it. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.
4. **Claim:** Socio-economic drought occurs when water shortage disrupts the supply of goods, services, livelihoods or essential uses; it expresses the interaction of physical scarcity with demand and unequal access. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.
5. **Claim:** Drought can unfold from rainfall deficit through soil-moisture stress, storage decline, ecosystem damage, livelihood loss and health effects over different time scales, so one indicator cannot represent every stage. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate,

uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.

6. **Claim:** India's Manual for Drought Management uses rainfall, vegetative indices, crop-sowing progression, soil moisture and hydrological indices in a graded assessment rather than one universal crop-loss percentage. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.
7. **Claim:** Forecast-triggered early action should precede severe impacts, but water-intensive cooling, indiscriminate groundwater extraction or unequal greening can increase future vulnerability and become maladaptation. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.

Counter-position / limit: An Act, plan, guideline, scheme, forecast, warning, drill, team, sanctioned capacity, allocation or dispatch cannot alone establish implementation, evacuation, expenditure, reduced loss, equitable recovery or resilience; test mandate, process, status and evidence.

Qualified conclusion: Claim: Meteorological drought. **Named evidence/example:** Meteorological drought concerns prolonged inadequate or poorly distributed precipitation relative to the relevant climatology; it is a hazard indicator, not a complete account of impacts. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Agricultural drought. **Named evidence/example:** Agricultural drought arises when soil moisture and crop-water availability become inadequate for crops, so rainfall information must be read with sowing, vegetation and soil-moisture evidence. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Hydrological drought. **Named evidence/example:** Hydrological drought concerns below-normal availability in rivers, reservoirs, lakes or aquifers and may lag behind the rainfall deficit that initiated it. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Socio-economic drought. **Named evidence/example:** Socio-economic drought occurs when water shortage disrupts the supply of goods, services, livelihoods or essential uses; it expresses the interaction of physical scarcity with demand and unequal access. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Slow-onset cascade. **Named evidence/example:** Drought can unfold from rainfall deficit through soil-moisture stress, storage decline, ecosystem damage, livelihood loss and health effects over different time scales, so one indicator cannot represent every stage. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Multi-indicator declaration. **Named evidence/example:** India's Manual for Drought Management uses rainfall, vegetative indices, crop-sowing progression, soil moisture and hydrological indices in a graded assessment rather than one universal crop-loss percentage. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Early action and maladaptation. **Named evidence/example:** Forecast-triggered early action should precede severe impacts, but water-intensive cooling, indiscriminate groundwater extraction or unequal greening can increase future vulnerability and become maladaptation. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

Executable exam-length answer / compression plan: For 15 marks, spend one-sixth of the time decoding the directive and drawing hazard -> exposure/vulnerability/capacity -> risk/impact -> prevention/mitigation/preparedness

-> response/recovery/BBB; write four to seven claim -> named evidence/example -> analysis -> qualification points; reserve the final minute for source, date, unit, status, mandate and event/inference checks.

Why this earns marks: The answer obeys the directive, uses India-centric evidence, and preserves risk terms, legal character, institutional mandate, warning/cycle stage, finance status, implementation and resilience distinctions.

How to improve this answer: For ‘Analyse drought as a slow-onset cascade requiring multi-indicator monitoring and early...', replace the weakest generic point with one exact Indian mechanism, institution/guideline/event, dated source and unit, implementation bottleneck, local-capacity or inclusion safeguard and answer-specific qualification.

ORIGINAL MAINS 4 - 15 MARKS

Question: Examine a Heat Action Plan through forecast, health, labour, water and urban-governance responsibilities. Answer in about 250 words.

Model thesis: Claim: Heat-wave hazard. **Named evidence/example:** A heat wave is a meteorological hazard identified through IMD's station and departure or actual-temperature criteria; it is not synonymous with every hot day or every episode of heat illness. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Heat stress. **Named evidence/example:** Heat stress is the physiological burden produced by temperature, humidity, radiation, wind, exertion, clothing, health and access to rest, water or cooling; exposure and vulnerability therefore mediate the hazard. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Exposure inequality. **Named evidence/example:** Outdoor and informal workers, older persons, children, people with illness, poorly housed households and those lacking dependable water or cooling may face different risk under the same forecast temperature. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Heat Action Plan. **Named evidence/example:** A Heat Action Plan links seasonal preparedness, warnings, health-system readiness, public communication, water and cooling access, work or school timing and longer-term urban measures through locally assigned responsibilities. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Urban heat measures. **Named evidence/example:** Cool roofs, shade, ventilation, trees and water bodies can reduce selected exposures, but benefits depend on design, maintenance, local climate and access; a cooling intervention can transfer costs or water demand. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Labour and health measures. **Named evidence/example:** Risk-sensitive work-rest scheduling, drinking water, shaded rest, worker communication, clinical readiness and surveillance are distinct from the meteorological forecast and require responsible employers and public agencies. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Monitoring-to-outcome firewall. **Named evidence/example:** A drought declaration, forecast, Heat Action Plan, dashboard, water scheme or contingency plan proves a classification or input; timely reach, equitable access, livelihood continuity and reduced illness require separate evidence. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

Claim -> named evidence -> analysis -> qualification:

- A heat wave is a meteorological hazard identified through IMD's station and departure or actual-temperature criteria; it is not synonymous with every hot day or every episode of heat illness.

- Heat stress is the physiological burden produced by temperature, humidity, radiation, wind, exertion, clothing, health and access to rest, water or cooling; exposure and vulnerability therefore mediate the hazard.
- Outdoor and informal workers, older persons, children, people with illness, poorly housed households and those lacking dependable water or cooling may face different risk under the same forecast temperature.
- A Heat Action Plan links seasonal preparedness, warnings, health-system readiness, public communication, water and cooling access, work or school timing and longer-term urban measures through locally assigned responsibilities.
- Cool roofs, shade, ventilation, trees and water bodies can reduce selected exposures, but benefits depend on design, maintenance, local climate and access; a cooling intervention can transfer costs or water demand.
- Risk-sensitive work-rest scheduling, drinking water, shaded rest, worker communication, clinical readiness and surveillance are distinct from the meteorological forecast and require responsible employers and public agencies.
- A drought declaration, forecast, Heat Action Plan, dashboard, water scheme or contingency plan proves a classification or input; timely reach, equitable access, livelihood continuity and reduced illness require separate evidence.

Qualified conclusion: Claim: Heat-wave hazard. **Named evidence/example:** A heat wave is a meteorological hazard identified through IMD's station and departure or actual-temperature criteria; it is not synonymous with every hot day or every episode of heat illness. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Heat stress. **Named evidence/example:** Heat stress is the physiological burden produced by temperature, humidity, radiation, wind, exertion, clothing, health and access to rest, water or cooling; exposure and vulnerability therefore mediate the hazard. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Exposure inequality. **Named evidence/example:** Outdoor and informal workers, older persons, children, people with illness, poorly housed households and those lacking dependable water or cooling may face different risk under the same forecast temperature. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Heat Action Plan. **Named evidence/example:** A Heat Action Plan links seasonal preparedness, warnings, health-system readiness, public communication, water and cooling access, work or school timing and longer-term urban measures through locally assigned responsibilities. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Urban heat measures. **Named evidence/example:** Cool roofs, shade, ventilation, trees and water bodies can reduce selected exposures, but benefits depend on design, maintenance, local climate and access; a cooling intervention can transfer costs or water demand. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Labour and health measures. **Named evidence/example:** Risk-sensitive work-rest scheduling, drinking water, shaded rest, worker communication, clinical readiness and surveillance are distinct from the meteorological forecast and require responsible employers and public agencies. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Monitoring-to-outcome firewall. **Named evidence/example:** A drought declaration, forecast, Heat Action Plan, dashboard, water scheme or contingency plan proves a classification or input; timely reach, equitable access, livelihood continuity and reduced illness require separate evidence. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

Demand decoding: The directive **examine** requires a direct position on ‘Examine a Heat Action Plan through forecast, health, labour, water and urban-governance...', each clause, risk mechanism, named Indian law/institution/event, dated status, implementation and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Heat-wave hazard. **Named evidence/example:** A heat wave is a meteorological hazard identified through IMD's station and departure or actual-temperature criteria; it is not synonymous with every hot day or every episode of heat illness. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Heat stress. **Named evidence/example:** Heat stress is the physiological burden produced by temperature, humidity, radiation, wind, exertion, clothing, health and access to rest, water or cooling; exposure and vulnerability therefore mediate the hazard. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Exposure inequality. **Named evidence/example:** Outdoor and informal workers, older persons, children, people with illness, poorly housed households and those lacking dependable water or cooling may face different risk under the same forecast temperature. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Heat Action Plan. **Named evidence/example:** A Heat Action Plan links seasonal preparedness, warnings, health-system readiness, public communication, water and cooling access, work or school timing and longer-term urban measures through locally assigned responsibilities. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Urban heat measures. **Named evidence/example:** Cool roofs, shade, ventilation, trees and water bodies can reduce selected exposures, but benefits depend on design, maintenance, local climate and access; a cooling intervention can transfer costs or water demand. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Labour and health measures. **Named evidence/example:** Risk-sensitive work-rest scheduling, drinking water, shaded rest, worker communication, clinical readiness and surveillance are distinct from the meteorological forecast and require responsible employers and public agencies. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Monitoring-to-outcome firewall. **Named evidence/example:** A drought declaration, forecast, Heat Action Plan, dashboard, water scheme or contingency plan proves a classification or input; timely reach, equitable access, livelihood continuity and reduced illness require separate evidence. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

Analytical body:

1. **Claim:** A heat wave is a meteorological hazard identified through IMD's station and departure or actual-temperature criteria; it is not synonymous with every hot day or every episode of heat illness. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.
2. **Claim:** Heat stress is the physiological burden produced by temperature, humidity, radiation, wind, exertion, clothing, health and access to rest, water or cooling; exposure and vulnerability therefore mediate the hazard. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.

3. **Claim:** Outdoor and informal workers, older persons, children, people with illness, poorly housed households and those lacking dependable water or cooling may face different risk under the same forecast temperature. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.
4. **Claim:** A Heat Action Plan links seasonal preparedness, warnings, health-system readiness, public communication, water and cooling access, work or school timing and longer-term urban measures through locally assigned responsibilities. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.
5. **Claim:** Cool roofs, shade, ventilation, trees and water bodies can reduce selected exposures, but benefits depend on design, maintenance, local climate and access; a cooling intervention can transfer costs or water demand. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.
6. **Claim:** Risk-sensitive work-rest scheduling, drinking water, shaded rest, worker communication, clinical readiness and surveillance are distinct from the meteorological forecast and require responsible employers and public agencies. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.
7. **Claim:** A drought declaration, forecast, Heat Action Plan, dashboard, water scheme or contingency plan proves a classification or input; timely reach, equitable access, livelihood continuity and reduced illness require separate evidence. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.

Counter-position / limit: An Act, plan, guideline, scheme, forecast, warning, drill, team, sanctioned capacity, allocation or dispatch cannot alone establish implementation, evacuation, expenditure, reduced loss, equitable recovery or resilience; test mandate, process, status and evidence.

Qualified conclusion: Claim: Heat-wave hazard. **Named evidence/example:** A heat wave is a meteorological hazard identified through IMD's station and departure or actual-temperature criteria; it is not synonymous with every hot day or every episode of heat illness. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Heat stress. **Named evidence/example:** Heat stress is the physiological burden produced by temperature, humidity, radiation, wind, exertion, clothing, health and access to rest, water or cooling; exposure and vulnerability therefore mediate the hazard. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Exposure inequality. **Named evidence/example:** Outdoor and informal workers, older persons, children, people with illness, poorly housed households and those lacking dependable water or cooling may face different risk under the same forecast temperature. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Heat Action Plan. **Named evidence/example:** A Heat Action Plan links seasonal preparedness, warnings, health-system readiness, public communication, water and cooling access, work or school timing and longer-term urban measures through locally assigned responsibilities. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Urban heat measures. **Named evidence/example:** Cool roofs, shade, ventilation, trees and water bodies can reduce selected exposures, but benefits depend on design, maintenance, local climate and access; a cooling intervention can transfer costs or water demand. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Labour and health measures. **Named evidence/example:** Risk-sensitive work-rest scheduling, drinking water, shaded rest, worker communication, clinical readiness and surveillance are distinct from the meteorological forecast and require responsible employers and public agencies. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Monitoring-to-outcome firewall. **Named evidence/example:** A drought declaration, forecast, Heat Action Plan, dashboard, water scheme or contingency plan proves a classification or input; timely reach, equitable access, livelihood continuity and reduced illness require separate evidence. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

Executable exam-length answer / compression plan: For 15 marks, spend one-sixth of the time decoding the directive and drawing hazard -> exposure/vulnerability/capacity -> risk/impact -> prevention/mitigation/preparedness -> response/recovery/BBB; write four to seven claim -> named evidence/example -> analysis -> qualification points; reserve the final minute for source, date, unit, status, mandate and event/inference checks.

Why this earns marks: The answer obeys the directive, uses India-centric evidence, and preserves risk terms, legal character, institutional mandate, warning/cycle stage, finance status, implementation and resilience distinctions.

How to improve this answer: For ‘Examine a Heat Action Plan through forecast, health, labour, water and urban-governance...', replace the weakest generic point with one exact Indian mechanism, institution/guideline/event, dated source and unit, implementation bottleneck, local-capacity or inclusion safeguard and answer-specific qualification.

ORIGINAL MAINS 5 - 20 MARKS

Question: Design an integrated drought-risk strategy for water, agriculture, livelihoods, health and social protection. Answer in about 300 words.

Model thesis: Claim: Aridity versus drought. **Named evidence/example:** Aridity is a long-term climatic condition of low moisture availability, whereas drought is a temporary or episodic deficit relative to the expected local water regime. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Meteorological drought. **Named evidence/example:** Meteorological drought concerns prolonged inadequate or poorly distributed precipitation relative to the relevant climatology; it is a hazard indicator, not a complete account of impacts. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Agricultural drought. **Named evidence/example:** Agricultural drought arises when soil moisture and crop-water availability become inadequate for crops, so rainfall information must be read with sowing, vegetation and soil-moisture evidence. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Hydrological drought. **Named evidence/example:** Hydrological drought concerns below-normal availability in rivers, reservoirs, lakes or aquifers and may lag behind the rainfall deficit that initiated it. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Socio-economic drought.

Named evidence/example: Socio-economic drought occurs when water shortage disrupts the supply of goods, services, livelihoods or essential uses; it expresses the interaction of physical scarcity with demand and unequal access. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Slow-onset cascade. **Named evidence/example:** Drought can unfold from rainfall deficit through soil-moisture stress, storage decline, ecosystem damage, livelihood loss and health effects over different time scales, so one indicator cannot represent every stage. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Multi-indicator declaration. **Named evidence/example:** India's Manual for Drought Management uses rainfall, vegetative indices, crop-sowing progression, soil moisture and hydrological indices in a graded assessment rather than one universal crop-loss percentage. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Water-risk portfolio. **Named evidence/example:** Drought management requires demand management, drinking-water security, groundwater and surface-water monitoring, watershed or recharge measures, storage governance and protection against shifting scarcity to weaker users. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Agriculture and livelihoods. **Named evidence/example:** Crop choice, contingency planning, soil-moisture conservation, livestock support, insurance or social protection and alternative livelihoods address agricultural and socioeconomic drought without treating relief as adaptation. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Compound risk. **Named evidence/example:** Drought, water scarcity, heat, wildfire risk, crop stress, power demand and public-health pressure can interact, while humid heat or hot nights may intensify harm without changing the formal hazard label. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Monitoring-to-outcome firewall. **Named evidence/example:** A drought declaration, forecast, Heat Action Plan, dashboard, water scheme or contingency plan proves a classification or input; timely reach, equitable access, livelihood continuity and reduced illness require separate evidence. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

Claim -> named evidence -> analysis -> qualification:

- Aridity is a long-term climatic condition of low moisture availability, whereas drought is a temporary or episodic deficit relative to the expected local water regime.
- Meteorological drought concerns prolonged inadequate or poorly distributed precipitation relative to the relevant climatology; it is a hazard indicator, not a complete account of impacts.
- Agricultural drought arises when soil moisture and crop-water availability become inadequate for crops, so rainfall information must be read with sowing, vegetation and soil-moisture evidence.
- Hydrological drought concerns below-normal availability in rivers, reservoirs, lakes or aquifers and may lag behind the rainfall deficit that initiated it.
- Socio-economic drought occurs when water shortage disrupts the supply of goods, services, livelihoods or essential uses; it expresses the interaction of physical scarcity with demand and unequal access.
- Drought can unfold from rainfall deficit through soil-moisture stress, storage decline, ecosystem damage, livelihood loss and health effects over different time scales, so one indicator cannot represent every stage.
- India's Manual for Drought Management uses rainfall, vegetative indices, crop-sowing progression, soil moisture and hydrological indices in a graded assessment rather than one universal crop-loss percentage.

- Drought management requires demand management, drinking-water security, groundwater and surface-water monitoring, watershed or recharge measures, storage governance and protection against shifting scarcity to weaker users.
- Crop choice, contingency planning, soil-moisture conservation, livestock support, insurance or social protection and alternative livelihoods address agricultural and socioeconomic drought without treating relief as adaptation.
- Drought, water scarcity, heat, wildfire risk, crop stress, power demand and public-health pressure can interact, while humid heat or hot nights may intensify harm without changing the formal hazard label.
- A drought declaration, forecast, Heat Action Plan, dashboard, water scheme or contingency plan proves a classification or input; timely reach, equitable access, livelihood continuity and reduced illness require separate evidence.

Qualified conclusion: Claim: Aridity versus drought. **Named evidence/example:** Aridity is a long-term climatic condition of low moisture availability, whereas drought is a temporary or episodic deficit relative to the expected local water regime. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Meteorological drought. **Named evidence/example:** Meteorological drought concerns prolonged inadequate or poorly distributed precipitation relative to the relevant climatology; it is a hazard indicator, not a complete account of impacts. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Agricultural drought. **Named evidence/example:** Agricultural drought arises when soil moisture and crop-water availability become inadequate for crops, so rainfall information must be read with sowing, vegetation and soil-moisture evidence. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Hydrological drought. **Named evidence/example:** Hydrological drought concerns below-normal availability in rivers, reservoirs, lakes or aquifers and may lag behind the rainfall deficit that initiated it. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Socio-economic drought. **Named evidence/example:** Socio-economic drought occurs when water shortage disrupts the supply of goods, services, livelihoods or essential uses; it expresses the interaction of physical scarcity with demand and unequal access. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Slow-onset cascade. **Named evidence/example:** Drought can unfold from rainfall deficit through soil-moisture stress, storage decline, ecosystem damage, livelihood loss and health effects over different time scales, so one indicator cannot represent every stage. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Multi-indicator declaration. **Named evidence/example:** India's Manual for Drought Management uses rainfall, vegetative indices, crop-sowing progression, soil moisture and hydrological indices in a graded assessment rather than one universal crop-loss percentage. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Water-risk portfolio. **Named evidence/example:** Drought management requires demand management, drinking-water security, groundwater and surface-water monitoring, watershed or recharge measures, storage governance and protection against shifting scarcity to weaker users. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Agriculture and livelihoods. **Named evidence/example:** Crop choice, contingency planning, soil-moisture conservation, livestock support, insurance or social protection and alternative livelihoods address agricultural and socioeconomic drought without treating relief as adaptation. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Compound risk. **Named evidence/example:** Drought, water scarcity, heat, wildfire risk, crop stress, power demand and public-health pressure can interact, while humid heat or hot nights may intensify harm without changing the formal hazard label. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Monitoring-to-outcome firewall. **Named evidence/example:** A drought declaration, forecast, Heat Action Plan, dashboard, water scheme or contingency plan proves a classification or input; timely reach, equitable access, livelihood continuity and reduced illness require separate evidence. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

Demand decoding: The directive **answer** requires a direct position on ‘Design an integrated drought-risk strategy for water, agriculture, livelihoods, health and...', each clause, risk mechanism, named Indian law/institution/event, dated status, implementation and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Aridity versus drought. **Named evidence/example:** Aridity is a long-term climatic condition of low moisture availability, whereas drought is a temporary or episodic deficit relative to the expected local water regime. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Meteorological drought. **Named evidence/example:** Meteorological drought concerns prolonged inadequate or poorly distributed precipitation relative to the relevant climatology; it is a hazard indicator, not a complete account of impacts. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Agricultural drought. **Named evidence/example:** Agricultural drought arises when soil moisture and crop-water availability become inadequate for crops, so rainfall information must be read with sowing, vegetation and soil-moisture evidence. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Hydrological drought. **Named evidence/example:** Hydrological drought concerns below-normal availability in rivers, reservoirs, lakes or aquifers and may lag behind the rainfall deficit that initiated it. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Socio-economic drought. **Named evidence/example:** Socio-economic drought occurs when water shortage disrupts the supply of goods, services, livelihoods or essential uses; it expresses the interaction of physical scarcity with demand and unequal access. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Slow-onset cascade. **Named evidence/example:** Drought can unfold from rainfall deficit through soil-moisture stress, storage decline, ecosystem damage, livelihood loss and health effects over different time scales, so one indicator cannot represent every stage. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Multi-indicator declaration. **Named evidence/example:** India's Manual for Drought Management uses rainfall, vegetative indices, crop-sowing progression, soil moisture and hydrological indices in a graded assessment rather than one universal crop-loss percentage. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Water-risk portfolio. **Named evidence/example:** Drought management requires demand management, drinking-water security, groundwater and surface-water monitoring, watershed or recharge measures, storage governance and protection against shifting scarcity to weaker users. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Agriculture and livelihoods. **Named evidence/example:** Crop choice, contingency planning, soil-moisture conservation, livestock support, insurance or social protection and alternative livelihoods address agricultural and socioeconomic drought without treating relief as adaptation. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Compound risk. **Named evidence/example:** Drought, water scarcity, heat, wildfire risk, crop stress, power demand and public-health pressure can interact, while humid heat or hot nights may intensify harm without changing the formal hazard label. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Monitoring-to-outcome firewall. **Named evidence/example:** A drought declaration, forecast, Heat Action Plan, dashboard, water scheme or contingency plan proves a classification or input; timely reach, equitable access, livelihood continuity and reduced illness require separate evidence. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

Analytical body:

1. **Claim:** Aridity is a long-term climatic condition of low moisture availability, whereas drought is a temporary or episodic deficit relative to the expected local water regime. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.
2. **Claim:** Meteorological drought concerns prolonged inadequate or poorly distributed precipitation relative to the relevant climatology; it is a hazard indicator, not a complete account of impacts. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.
3. **Claim:** Agricultural drought arises when soil moisture and crop-water availability become inadequate for crops, so rainfall information must be read with sowing, vegetation and soil-moisture evidence. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.
4. **Claim:** Hydrological drought concerns below-normal availability in rivers, reservoirs, lakes or aquifers and may lag behind the rainfall deficit that initiated it. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.
5. **Claim:** Socio-economic drought occurs when water shortage disrupts the supply of goods, services, livelihoods or essential uses; it expresses the interaction of physical scarcity with demand and unequal access. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.
6. **Claim:** Drought can unfold from rainfall deficit through soil-moisture stress, storage decline, ecosystem damage, livelihood loss and health effects over different time scales, so one indicator cannot represent every stage. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence ->

competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.

7. **Claim:** India's Manual for Drought Management uses rainfall, vegetative indices, crop-sowing progression, soil moisture and hydrological indices in a graded assessment rather than one universal crop-loss percentage. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.
8. **Claim:** Drought management requires demand management, drinking-water security, groundwater and surface-water monitoring, watershed or recharge measures, storage governance and protection against shifting scarcity to weaker users. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.
9. **Claim:** Crop choice, contingency planning, soil-moisture conservation, livestock support, insurance or social protection and alternative livelihoods address agricultural and socioeconomic drought without treating relief as adaptation. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.
10. **Claim:** Drought, water scarcity, heat, wildfire risk, crop stress, power demand and public-health pressure can interact, while humid heat or hot nights may intensify harm without changing the formal hazard label. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.
11. **Claim:** A drought declaration, forecast, Heat Action Plan, dashboard, water scheme or contingency plan proves a classification or input; timely reach, equitable access, livelihood continuity and reduced illness require separate evidence. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.

Counter-position / limit: An Act, plan, guideline, scheme, forecast, warning, drill, team, sanctioned capacity, allocation or dispatch cannot alone establish implementation, evacuation, expenditure, reduced loss, equitable recovery or resilience; test mandate, process, status and evidence.

Qualified conclusion: Claim: Aridity versus drought. **Named evidence/example:** Aridity is a long-term climatic condition of low moisture availability, whereas drought is a temporary or episodic deficit relative to the expected local water regime. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Meteorological drought. **Named evidence/example:** Meteorological drought concerns prolonged inadequate or poorly distributed precipitation relative to the relevant climatology; it is a hazard indicator, not a complete account of impacts. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Agricultural drought. **Named evidence/example:** Agricultural drought arises when soil moisture and crop-water availability become inadequate for crops, so rainfall information must be read with sowing, vegetation and soil-moisture evidence. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Hydrological drought. **Named evidence/example:** Hydrological drought concerns below-normal availability in rivers, reservoirs, lakes or aquifers and may lag behind the rainfall deficit that initiated it. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Socio-economic drought. **Named evidence/example:** Socio-economic drought occurs when water shortage disrupts the supply of goods, services, livelihoods or essential uses; it expresses the interaction of physical scarcity with demand and unequal access. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Slow-onset cascade. **Named evidence/example:** Drought can unfold from rainfall deficit through soil-moisture stress, storage decline, ecosystem damage, livelihood loss and health effects over different time scales, so one indicator cannot represent every stage. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Multi-indicator declaration. **Named evidence/example:** India's Manual for Drought Management uses rainfall, vegetative indices, crop-sowing progression, soil moisture and hydrological indices in a graded assessment rather than one universal crop-loss percentage. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Water-risk portfolio. **Named evidence/example:** Drought management requires demand management, drinking-water security, groundwater and surface-water monitoring, watershed or recharge measures, storage governance and protection against shifting scarcity to weaker users. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Agriculture and livelihoods. **Named evidence/example:** Crop choice, contingency planning, soil-moisture conservation, livestock support, insurance or social protection and alternative livelihoods address agricultural and socioeconomic drought without treating relief as adaptation. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Compound risk. **Named evidence/example:** Drought, water scarcity, heat, wildfire risk, crop stress, power demand and public-health pressure can interact, while humid heat or hot nights may intensify harm without changing the formal hazard label. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Monitoring-to-outcome firewall. **Named evidence/example:** A drought declaration, forecast, Heat Action Plan, dashboard, water scheme or contingency plan proves a classification or input; timely reach, equitable access, livelihood continuity and reduced illness require separate evidence. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

Executable exam-length answer / compression plan: For 20 marks, spend one-sixth of the time decoding the directive and drawing hazard -> exposure/vulnerability/capacity -> risk/impact -> prevention/mitigation/preparedness -> response/recovery/BBB; write four to seven claim -> named evidence/example -> analysis -> qualification points; reserve the final minute for source, date, unit, status, mandate and event/inference checks.

Why this earns marks: The answer obeys the directive, uses India-centric evidence, and preserves risk terms, legal character, institutional mandate, warning/cycle stage, finance status, implementation and resilience distinctions.

How to improve this answer: For ‘Design an integrated drought-risk strategy for water, agriculture, livelihoods, health and...', replace the weakest generic point with one exact Indian mechanism, institution/guideline/event, dated source and unit, implementation bottleneck, local-capacity or inclusion safeguard and answer-specific qualification.

ORIGINAL MAINS 6 - 20 MARKS

Question: Critically evaluate compound drought-heat risk, anticipatory action and maladaptation in Indian cities and rural regions. Answer in about 300 words.

Model thesis: Claim: Slow-onset cascade. **Named evidence/example:** Drought can unfold from rainfall deficit through soil-moisture stress, storage decline, ecosystem damage, livelihood loss and health effects over different time scales, so one indicator cannot represent every stage. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Heat stress. **Named evidence/example:** Heat stress is the physiological burden produced by temperature, humidity, radiation, wind, exertion, clothing, health and access to rest, water or cooling; exposure and vulnerability therefore mediate the hazard. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Wet-bulb boundary. **Named evidence/example:** Wet-bulb temperature combines heat and humidity by representing evaporative-cooling potential; it is not an IMD heat-wave declaration threshold, and dangerous stress can occur below a theoretical upper-limit benchmark. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Exposure inequality. **Named evidence/example:** Outdoor and informal workers, older persons, children, people with illness, poorly housed households and those lacking dependable water or cooling may face different risk under the same forecast temperature. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Heat Action Plan. **Named evidence/example:** A Heat Action Plan links seasonal preparedness, warnings, health-system readiness, public communication, water and cooling access, work or school timing and longer-term urban measures through locally assigned responsibilities. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Water-risk portfolio. **Named evidence/example:** Drought management requires demand management, drinking-water security, groundwater and surface-water monitoring, watershed or recharge measures, storage governance and protection against shifting scarcity to weaker users. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Urban heat measures. **Named evidence/example:** Cool roofs, shade, ventilation, trees and water bodies can reduce selected exposures, but benefits depend on design, maintenance, local climate and access; a cooling intervention can transfer costs or water demand. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Labour and health measures. **Named evidence/example:** Risk-sensitive work-rest scheduling, drinking water, shaded rest, worker communication, clinical readiness and surveillance are distinct from the meteorological forecast and require responsible employers and public agencies. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Compound risk. **Named evidence/example:** Drought, water scarcity, heat, wildfire risk, crop stress, power demand and public-health pressure can interact, while humid heat or hot nights may intensify harm without changing the formal hazard label. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Early action and maladaptation. **Named evidence/example:** Forecast-triggered early action should precede severe impacts, but water-intensive cooling, indiscriminate groundwater extraction or unequal greening can increase future vulnerability and become maladaptation. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Monitoring-to-outcome firewall. **Named evidence/example:** A drought declaration, forecast, Heat Action Plan, dashboard, water scheme or contingency plan proves a classification or input; timely reach, equitable access, livelihood continuity and reduced illness require separate evidence. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

Claim -> named evidence -> analysis -> qualification:

- Drought can unfold from rainfall deficit through soil-moisture stress, storage decline, ecosystem damage, livelihood loss and health effects over different time scales, so one indicator cannot represent every stage.
- Heat stress is the physiological burden produced by temperature, humidity, radiation, wind, exertion, clothing, health and access to rest, water or cooling; exposure and vulnerability therefore mediate the hazard.
- Wet-bulb temperature combines heat and humidity by representing evaporative-cooling potential; it is not an IMD heat-wave declaration threshold, and dangerous stress can occur below a theoretical upper-limit benchmark.
- Outdoor and informal workers, older persons, children, people with illness, poorly housed households and those lacking dependable water or cooling may face different risk under the same forecast temperature.
- A Heat Action Plan links seasonal preparedness, warnings, health-system readiness, public communication, water and cooling access, work or school timing and longer-term urban measures through locally assigned responsibilities.
- Drought management requires demand management, drinking-water security, groundwater and surface-water monitoring, watershed or recharge measures, storage governance and protection against shifting scarcity to weaker users.
- Cool roofs, shade, ventilation, trees and water bodies can reduce selected exposures, but benefits depend on design, maintenance, local climate and access; a cooling intervention can transfer costs or water demand.
- Risk-sensitive work-rest scheduling, drinking water, shaded rest, worker communication, clinical readiness and surveillance are distinct from the meteorological forecast and require responsible employers and public agencies.
- Drought, water scarcity, heat, wildfire risk, crop stress, power demand and public-health pressure can interact, while humid heat or hot nights may intensify harm without changing the formal hazard label.
- Forecast-triggered early action should precede severe impacts, but water-intensive cooling, indiscriminate groundwater extraction or unequal greening can increase future vulnerability and become maladaptation.
- A drought declaration, forecast, Heat Action Plan, dashboard, water scheme or contingency plan proves a classification or input; timely reach, equitable access, livelihood continuity and reduced illness require separate evidence.

Qualified conclusion: Claim: Slow-onset cascade. **Named evidence/example:** Drought can unfold from rainfall deficit through soil-moisture stress, storage decline, ecosystem damage, livelihood loss and health effects over different time scales, so one indicator cannot represent every stage. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Heat stress. **Named evidence/example:** Heat stress is the physiological burden produced by temperature, humidity, radiation, wind, exertion, clothing, health and access to rest, water or cooling; exposure and vulnerability therefore mediate the hazard. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Wet-bulb boundary. **Named evidence/example:** Wet-bulb temperature combines heat and humidity by representing evaporative-cooling potential; it is not an IMD heat-wave declaration threshold, and dangerous stress can occur below a theoretical upper-limit benchmark. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Exposure inequality. **Named evidence/example:** Outdoor and informal workers, older persons, children, people with illness, poorly housed households and those lacking dependable water or cooling may face different risk under the same forecast temperature. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Heat Action Plan. **Named evidence/example:** A Heat Action Plan links seasonal preparedness, warnings, health-system readiness, public communication, water and cooling access, work or school timing and longer-term urban measures through locally assigned responsibilities. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Water-risk portfolio. **Named evidence/example:** Drought management requires demand management, drinking-water security, groundwater and surface-water monitoring, watershed or recharge measures, storage governance and protection against shifting scarcity to weaker users. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Urban heat measures. **Named evidence/example:** Cool roofs, shade, ventilation, trees and water bodies can reduce selected exposures, but benefits depend on design, maintenance, local climate and access; a cooling intervention can transfer costs or water demand. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Labour and health measures. **Named evidence/example:** Risk-sensitive work-rest scheduling, drinking water, shaded rest, worker communication, clinical readiness and surveillance are distinct from the meteorological forecast and require responsible employers and public agencies. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Compound risk. **Named evidence/example:** Drought, water scarcity, heat, wildfire risk, crop stress, power demand and public-health pressure can interact, while humid heat or hot nights may intensify harm without changing the formal hazard label. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Early action and maladaptation. **Named evidence/example:** Forecast-triggered early action should precede severe impacts, but water-intensive cooling, indiscriminate groundwater extraction or unequal greening can increase future vulnerability and become maladaptation. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Monitoring-to-outcome firewall. **Named evidence/example:** A drought declaration, forecast, Heat Action Plan, dashboard, water scheme or contingency plan proves a classification or input; timely reach, equitable access, livelihood continuity and reduced illness require separate evidence. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

Demand decoding: The directive **evaluate** requires a direct position on ‘Critically evaluate compound drought-heat risk, anticipatory action and maladaptation in...', each clause, risk mechanism, named Indian law/institution/event, dated status, implementation and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Slow-onset cascade. **Named evidence/example:** Drought can unfold from rainfall deficit through soil-moisture stress, storage decline, ecosystem damage, livelihood loss and health effects over different time scales, so one indicator cannot represent every stage. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Heat stress. **Named evidence/example:** Heat stress is the physiological burden produced by temperature, humidity, radiation, wind, exertion, clothing, health and access to rest, water or cooling; exposure and vulnerability therefore mediate the hazard. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Wet-bulb boundary. **Named evidence/example:** Wet-bulb temperature combines heat and humidity by representing evaporative-cooling potential; it is not an IMD heat-wave declaration threshold, and dangerous stress can occur below a theoretical upper-limit benchmark. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Exposure inequality. **Named evidence/example:** Outdoor and informal workers, older persons, children, people with illness, poorly housed households and those lacking dependable water or cooling may face different risk under the same forecast temperature. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Heat Action Plan. **Named evidence/example:** A Heat Action Plan links seasonal preparedness, warnings, health-system readiness, public communication, water and cooling access, work or school timing and longer-term urban measures through locally assigned responsibilities. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Water-risk portfolio. **Named evidence/example:** Drought management requires demand management, drinking-water security, groundwater and surface-water monitoring, watershed or recharge measures, storage governance and protection against shifting scarcity to weaker users. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Urban heat measures. **Named evidence/example:** Cool roofs, shade, ventilation, trees and water bodies can reduce selected exposures, but benefits depend on design, maintenance, local climate and access; a cooling intervention can transfer costs or water demand. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Labour and health measures. **Named evidence/example:** Risk-sensitive work-rest scheduling, drinking water, shaded rest, worker communication, clinical readiness and surveillance are distinct from the meteorological forecast and require responsible employers and public agencies. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Compound risk. **Named evidence/example:** Drought, water scarcity, heat, wildfire risk, crop stress, power demand and public-health pressure can interact, while humid heat or hot nights may intensify harm without changing the formal hazard label. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Early action and maladaptation. **Named evidence/example:** Forecast-triggered early action should precede severe impacts, but water-intensive cooling, indiscriminate groundwater extraction or unequal greening can increase future vulnerability and become maladaptation. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Monitoring-to-outcome firewall. **Named evidence/example:** A drought declaration, forecast, Heat Action Plan, dashboard, water scheme or contingency plan proves a classification or input; timely reach, equitable access, livelihood continuity and reduced illness require separate evidence. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

Analytical body:

1. **Claim:** Drought can unfold from rainfall deficit through soil-moisture stress, storage decline, ecosystem damage, livelihood loss and health effects over different time scales, so one indicator cannot represent every stage. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.
2. **Claim:** Heat stress is the physiological burden produced by temperature, humidity, radiation, wind, exertion, clothing, health and access to rest, water or cooling; exposure and vulnerability therefore mediate the hazard. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or

dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.

3. **Claim:** Wet-bulb temperature combines heat and humidity by representing evaporative-cooling potential; it is not an IMD heat-wave declaration threshold, and dangerous stress can occur below a theoretical upper-limit benchmark. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.
4. **Claim:** Outdoor and informal workers, older persons, children, people with illness, poorly housed households and those lacking dependable water or cooling may face different risk under the same forecast temperature. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.
5. **Claim:** A Heat Action Plan links seasonal preparedness, warnings, health-system readiness, public communication, water and cooling access, work or school timing and longer-term urban measures through locally assigned responsibilities. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.
6. **Claim:** Drought management requires demand management, drinking-water security, groundwater and surface-water monitoring, watershed or recharge measures, storage governance and protection against shifting scarcity to weaker users. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.
7. **Claim:** Cool roofs, shade, ventilation, trees and water bodies can reduce selected exposures, but benefits depend on design, maintenance, local climate and access; a cooling intervention can transfer costs or water demand. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.
8. **Claim:** Risk-sensitive work-rest scheduling, drinking water, shaded rest, worker communication, clinical readiness and surveillance are distinct from the meteorological forecast and require responsible employers and public agencies. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.
9. **Claim:** Drought, water scarcity, heat, wildfire risk, crop stress, power demand and public-health pressure can interact, while humid heat or hot nights may intensify harm without changing the formal hazard label. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.
10. **Claim:** Forecast-triggered early action should precede severe impacts, but water-intensive cooling, indiscriminate groundwater extraction or unequal greening can increase future vulnerability and become maladaptation. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity ->

disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.

11. **Claim:** A drought declaration, forecast, Heat Action Plan, dashboard, water scheme or contingency plan proves a classification or input; timely reach, equitable access, livelihood continuity and reduced illness require separate evidence. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.

Counter-position / limit: An Act, plan, guideline, scheme, forecast, warning, drill, team, sanctioned capacity, allocation or dispatch cannot alone establish implementation, evacuation, expenditure, reduced loss, equitable recovery or resilience; test mandate, process, status and evidence.

Qualified conclusion: Claim: Slow-onset cascade. **Named evidence/example:** Drought can unfold from rainfall deficit through soil-moisture stress, storage decline, ecosystem damage, livelihood loss and health effects over different time scales, so one indicator cannot represent every stage. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Heat stress. **Named evidence/example:** Heat stress is the physiological burden produced by temperature, humidity, radiation, wind, exertion, clothing, health and access to rest, water or cooling; exposure and vulnerability therefore mediate the hazard. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Wet-bulb boundary. **Named evidence/example:** Wet-bulb temperature combines heat and humidity by representing evaporative-cooling potential; it is not an IMD heat-wave declaration threshold, and dangerous stress can occur below a theoretical upper-limit benchmark. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Exposure inequality. **Named evidence/example:** Outdoor and informal workers, older persons, children, people with illness, poorly housed households and those lacking dependable water or cooling may face different risk under the same forecast temperature. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Heat Action Plan. **Named evidence/example:** A Heat Action Plan links seasonal preparedness, warnings, health-system readiness, public communication, water and cooling access, work or school timing and longer-term urban measures through locally assigned responsibilities. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Water-risk portfolio. **Named evidence/example:** Drought management requires demand management, drinking-water security, groundwater and surface-water monitoring, watershed or recharge measures, storage governance and protection against shifting scarcity to weaker users. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Urban heat measures. **Named evidence/example:** Cool roofs, shade, ventilation, trees and water bodies can reduce selected exposures, but benefits depend on design, maintenance, local climate and access; a cooling intervention can transfer costs or water demand. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Labour and health measures. **Named evidence/example:** Risk-sensitive work-rest scheduling, drinking water, shaded rest, worker communication, clinical readiness and surveillance are distinct from the meteorological forecast and require responsible employers and public agencies. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Compound risk. **Named evidence/example:** Drought, water scarcity, heat, wildfire risk, crop stress, power demand and public-health pressure can interact, while humid heat or hot nights may intensify harm without changing the formal hazard label. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Early action and maladaptation. **Named evidence/example:** Forecast-triggered early action should precede severe impacts, but water-intensive cooling, indiscriminate groundwater extraction or unequal greening can increase future vulnerability and become maladaptation. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Monitoring-to-outcome firewall. **Named evidence/example:** A drought declaration, forecast, Heat Action Plan, dashboard, water scheme or contingency plan proves a classification or input; timely reach, equitable access, livelihood continuity and reduced illness require separate evidence. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

Executable exam-length answer / compression plan: For 20 marks, spend one-sixth of the time decoding the directive and drawing hazard -> exposure/vulnerability/capacity -> risk/impact -> prevention/mitigation/preparedness -> response/recovery/BBB; write four to seven claim -> named evidence/example -> analysis -> qualification points; reserve the final minute for source, date, unit, status, mandate and event/inference checks.

Why this earns marks: The answer obeys the directive, uses India-centric evidence, and preserves risk terms, legal character, institutional mandate, warning/cycle stage, finance status, implementation and resilience distinctions.

How to improve this answer: For ‘Critically evaluate compound drought-heat risk, anticipatory action and maladaptation in...', replace the weakest generic point with one exact Indian mechanism, institution/guideline/event, dated source and unit, implementation bottleneck, local-capacity or inclusion safeguard and answer-specific qualification.